

ECSE 4560/ 6560

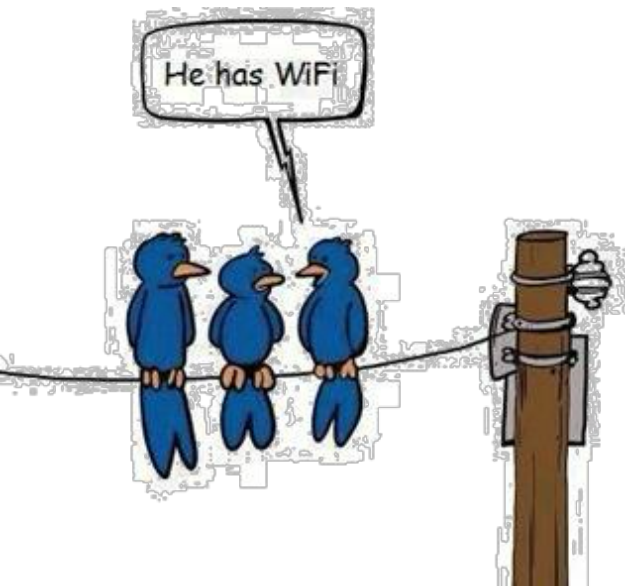
Modern Communication Systems

Lecture 24: Wireless Sensing Part 5:

FMCW Imaging

Ish Jain, Fall 2025

Camera / Lidar



Slides from Fadel, Mingmin, Dina Katabi

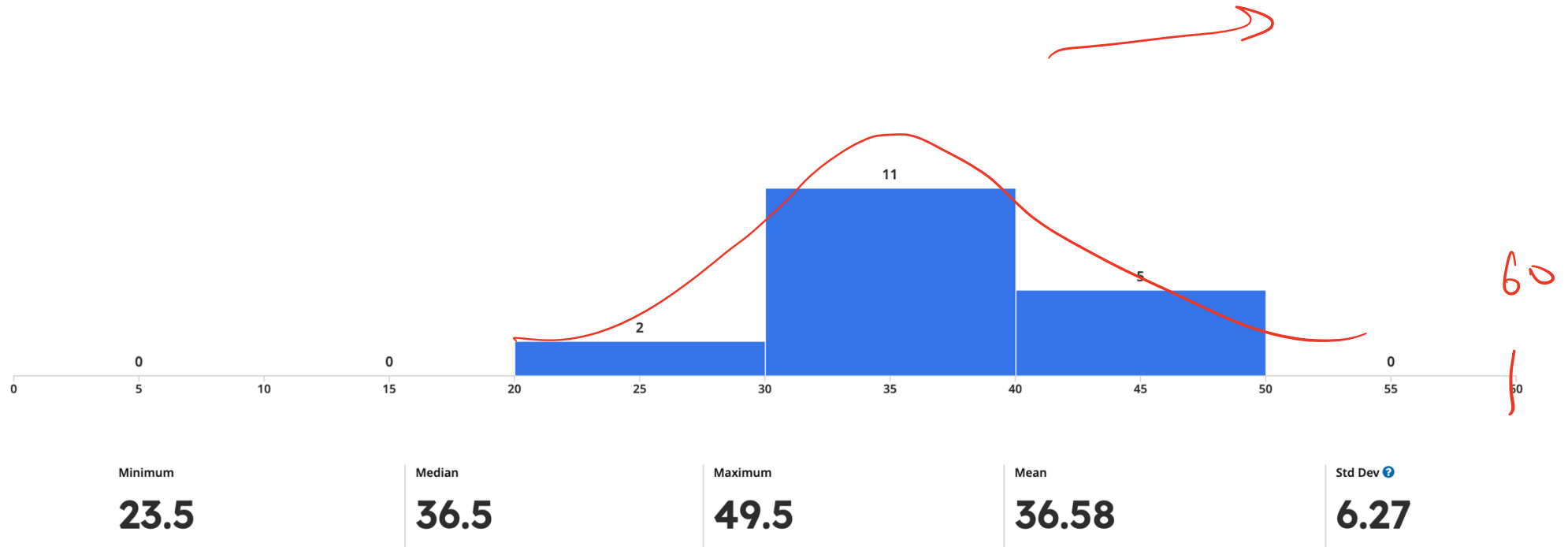
Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **FMCW Imaging**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
 - Final Project Presentation and Demos (5 min each group): **Fri 12/12**
 - Project Report and code submission: **Mon 12/15**
 - Peer evaluation (evaluate peers report and code): **Fri 12/19**
 - Instructions to be posted on Piazza
- HW/Exams Dates
 - HW 6, HW 7 and Exam2 will be graded today
 - HW 8 will be posted today (optional); due **Fri 12/12**
 - Best 7 of 8 HWs will be counted towards final grade.
 - Bottom 2 Quizzes will be dropped

in-class

online

Midterm2 Stats



feedback

Today's Class

- **Wireless Sensing – FMCW Imaging**

- RangeFFT *← revision*

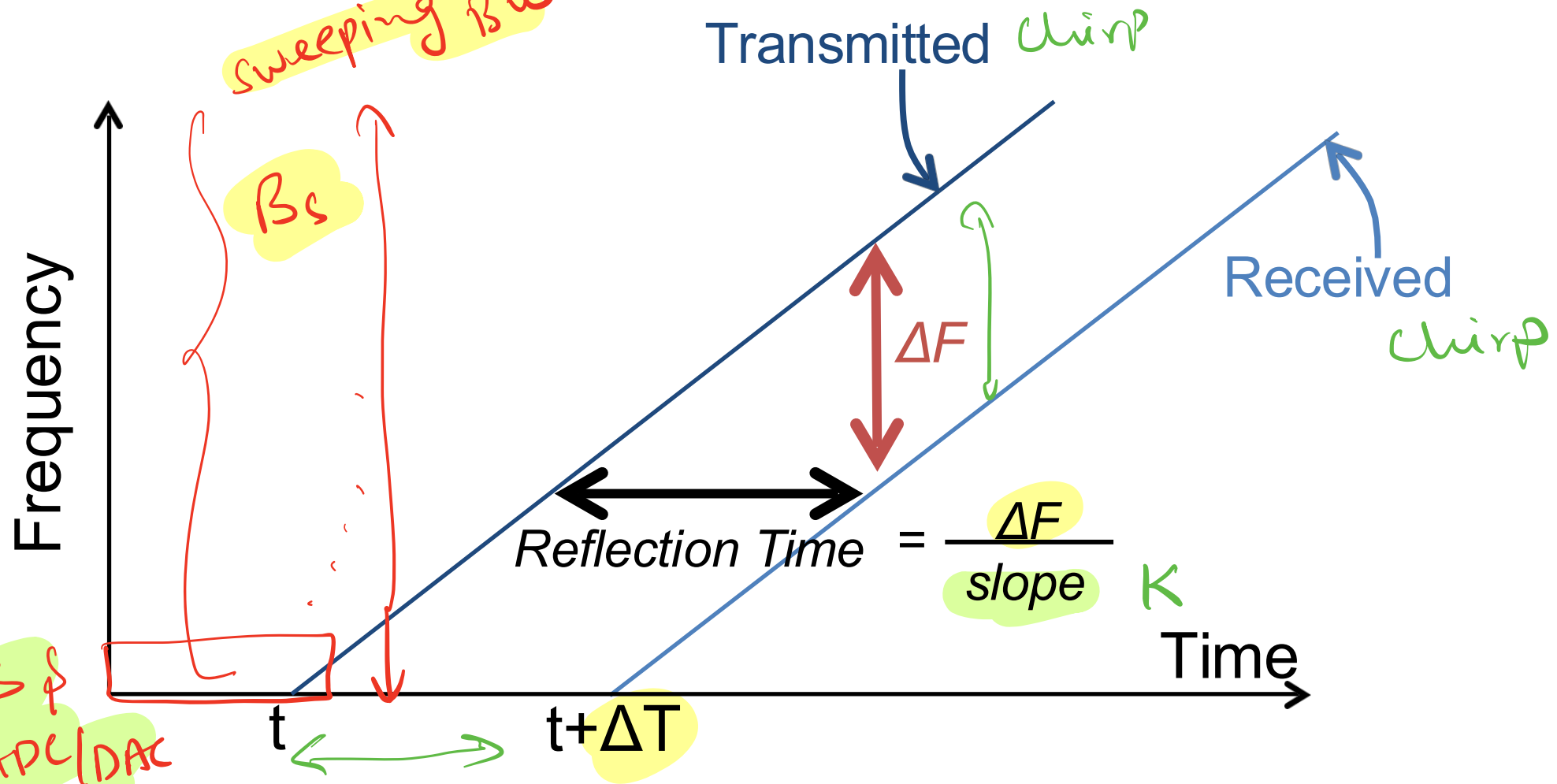
- Range Resolution in FMCW Radar

- Range-AoA 2D FFT → Imaging

- Doppler-FFT

*Breathing / HR
detection
Imaging improve*

FMCW: Measure time by measuring frequency

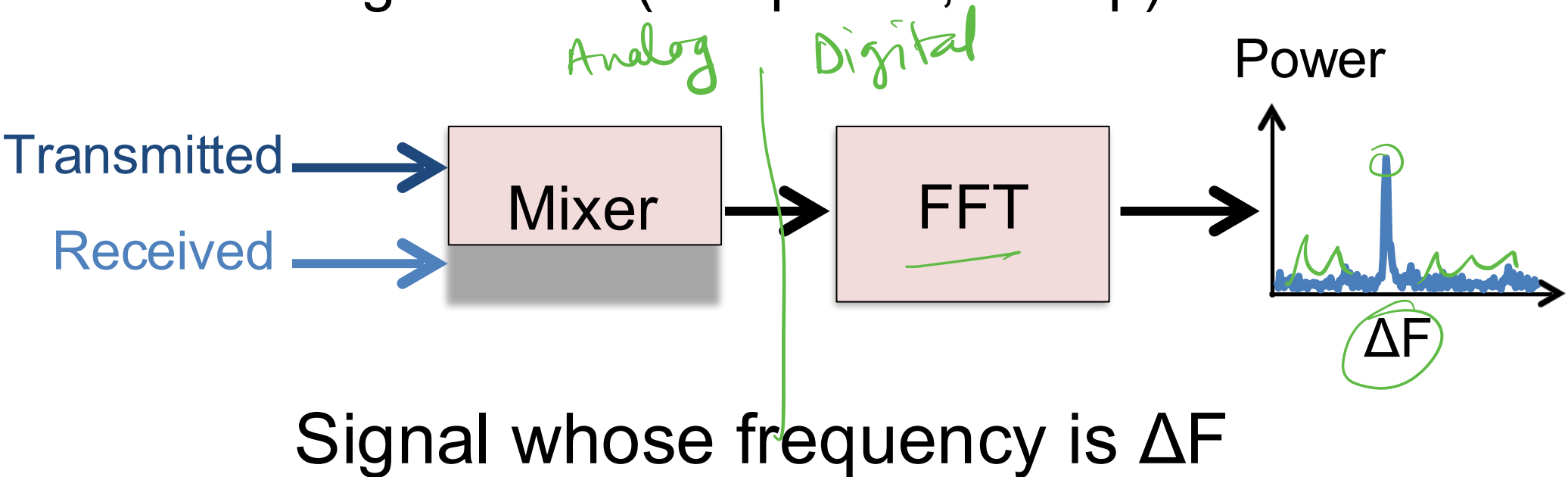


How do we measure ΔF ?

$$\Delta T = \Delta f / K$$

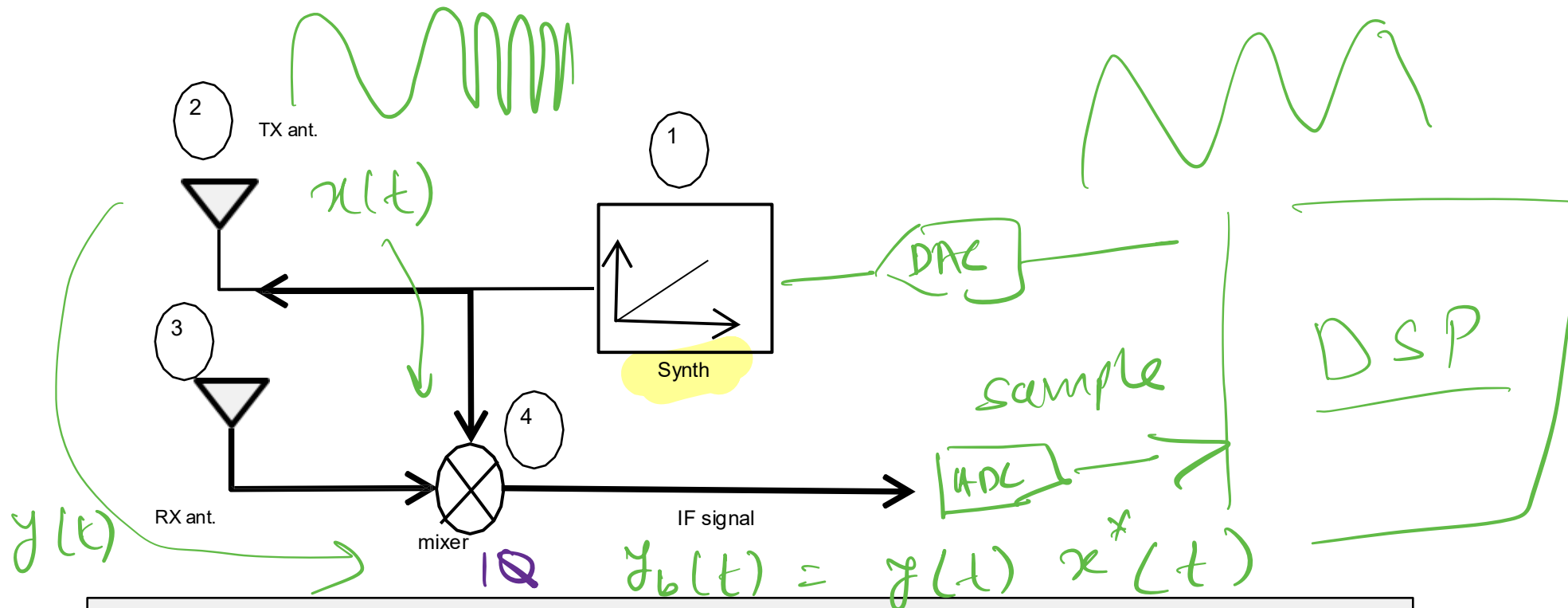
Measuring ΔF

- Subtracting frequencies is easy (e.g., removing carrier in WiFi)
- Done using a mixer (low-power; cheap)



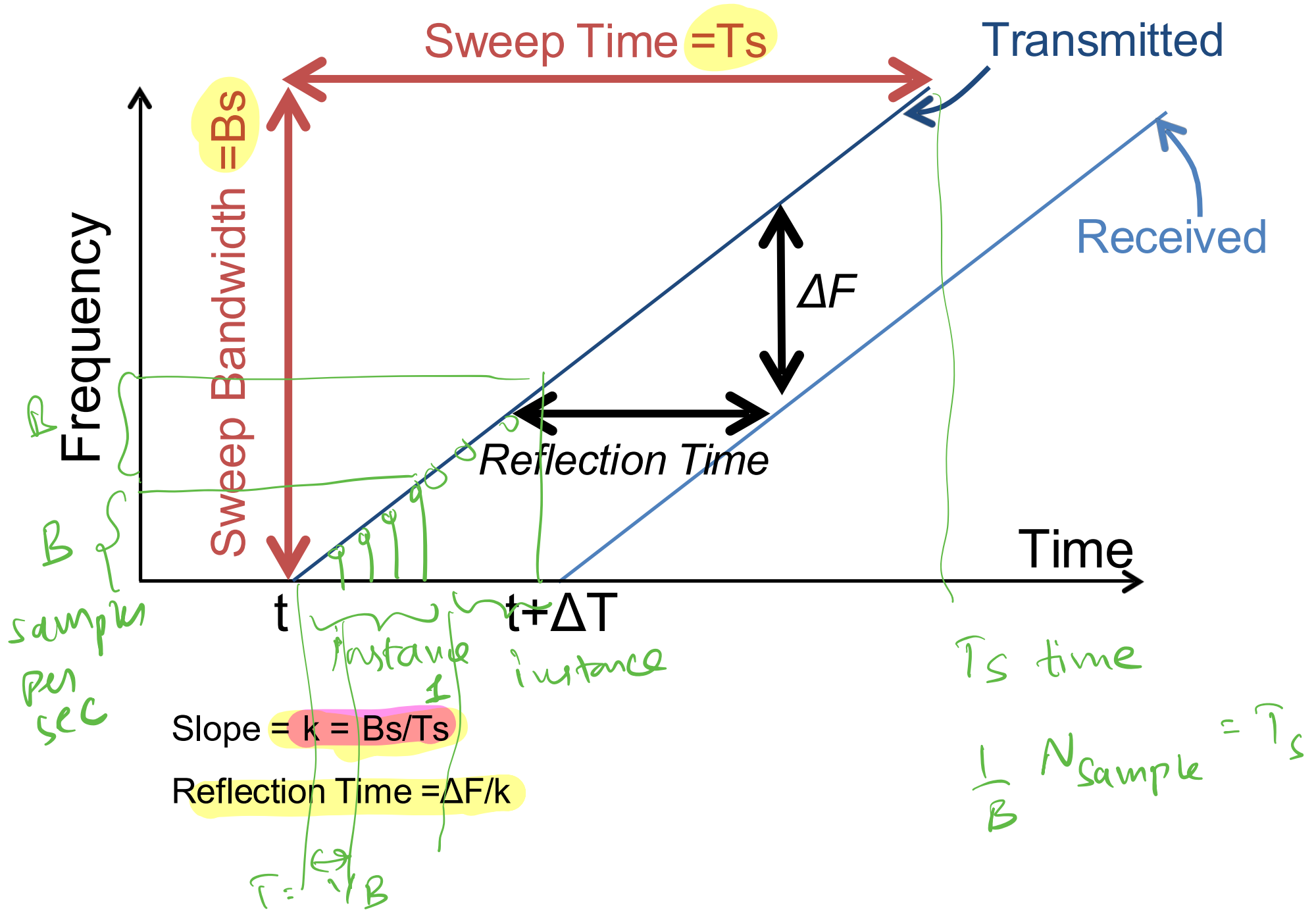
$\Delta F \rightarrow$ Reflection Time $\rightarrow$ Distance

FMCW Radar Architecture



1. A synthesizer (synth) generates a chirp
2. The chirp is transmitted by the TX antenna
3. The chirp is reflected off an object and the reflected chirp is received at the RX antenna.
4. The RX signal and TX signal are 'mixed' and the resulting signal is called an 'IF signal'. We'll analyze the IF signal in more detail in the next slide

FMCW: Measure time by measuring frequency



FMCW

- FMCW Transmitted Signal

$$x(t) = e^{j2\pi(\frac{k}{2}(t^2 + f_0 t))}$$

Frequency is linear in time;
hence phase is quadratic

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2}((t-\tau_i)^2 + f_0(t-\tau_i)))}$$

reflector i

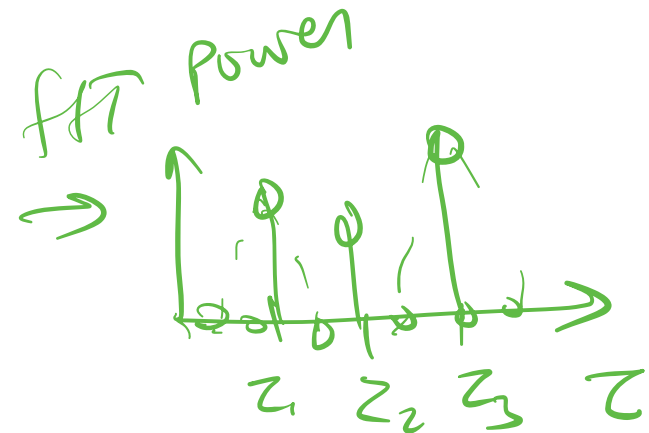
delay τ_i

Reflections linearly
combine over the
wireless medium

Conversion:

$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$

frequency $k\tau_i$



FMCW

- FMCW Transmitted Signal:

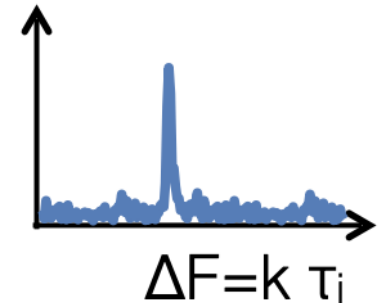
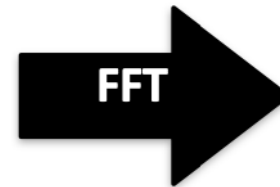
$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FMCW after downconversion:

$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$





FMCW Max Range

(to avoid ambiguity)

$\Delta F =$

δ
 $B + \delta$
 $2B + \delta$
 same
 can't distinguish

- FMCW Transmitted Signal:

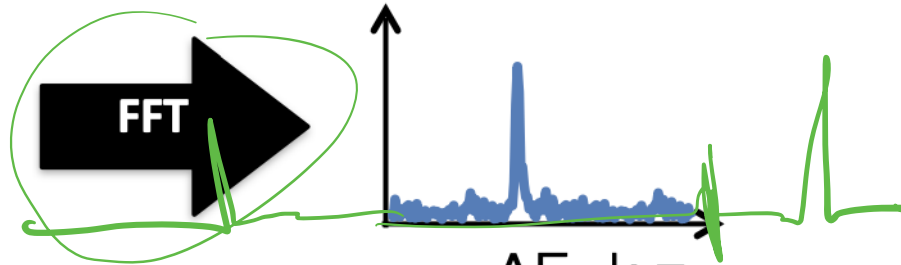
$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FMCW after downconversion:

$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$



- Sampling Rate = B

$$\Delta F < B$$

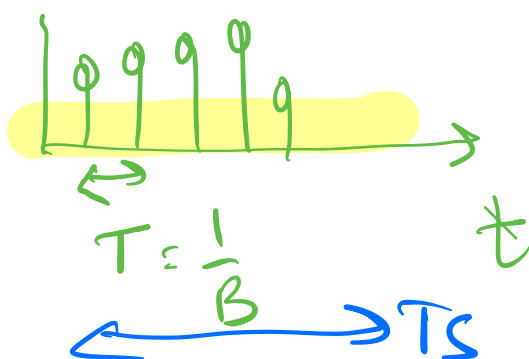
$$\rightarrow T_{\max} = B/k$$

$$= B \times T_s / B_s$$

$$d_{\max} = \frac{\Delta F = k \tau_i}{2 \times \text{max}} \times c/2$$

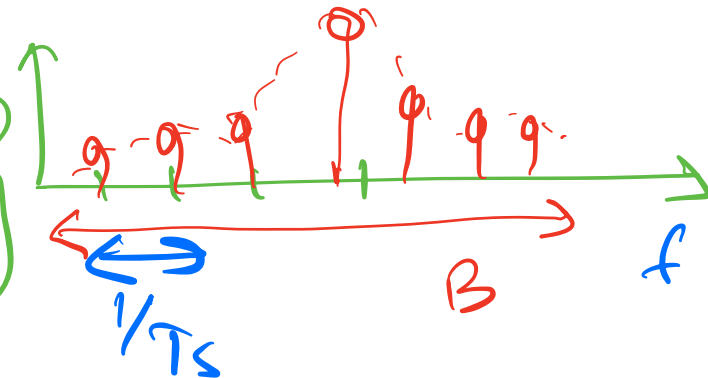
$$d_{\max} = \frac{c \times B \times T_s}{2 B_s}$$

$y_b(t)$



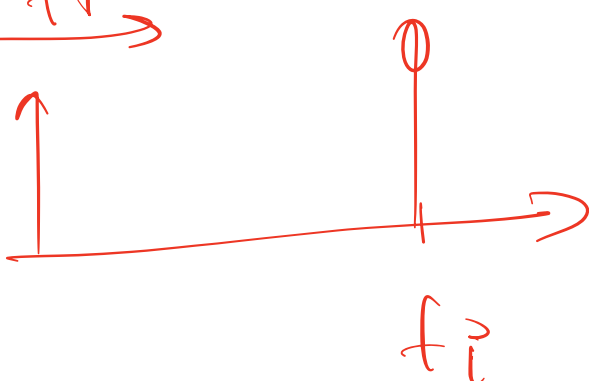
FFT

FFT
 $(y_b(t))$
 $= Y_b(f)$



$$y_b(t) \propto e^{j2\pi f_i t}$$

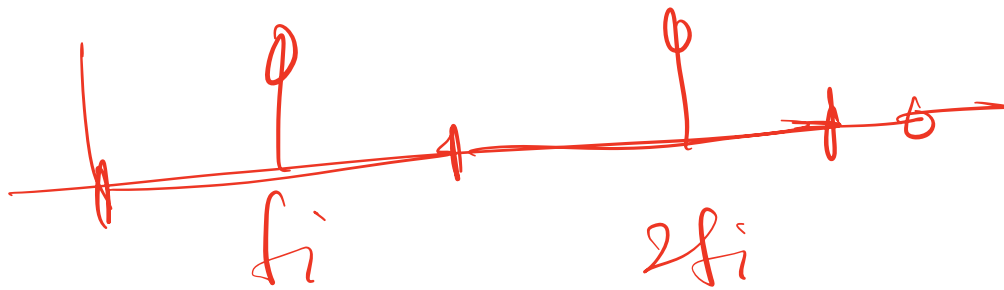
$f_i = k z_i$

$\xrightarrow{f_i}$


$$t = nT$$

$$T = \frac{1}{B}$$

$$y_b(nT) \propto e^{j2\pi f_i nT} \rightarrow$$

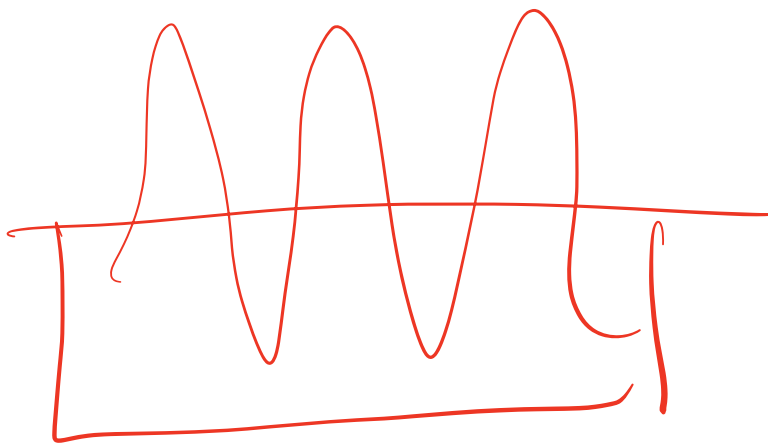


range of $n \rightarrow 0$ to N_{chirp}

Chirp duration $\geq T_S$

$$\Rightarrow N_{\text{samples}} / \text{Chirp} = \frac{T_S}{T}$$

$$e^{j2\pi f t}$$



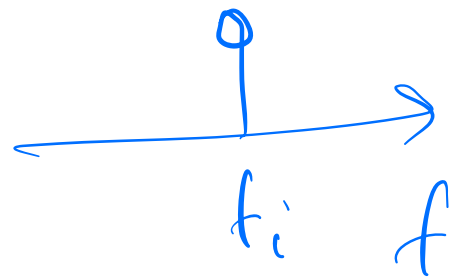
T_s

$$\text{or } N_s = \frac{T_s}{T}$$

samplers

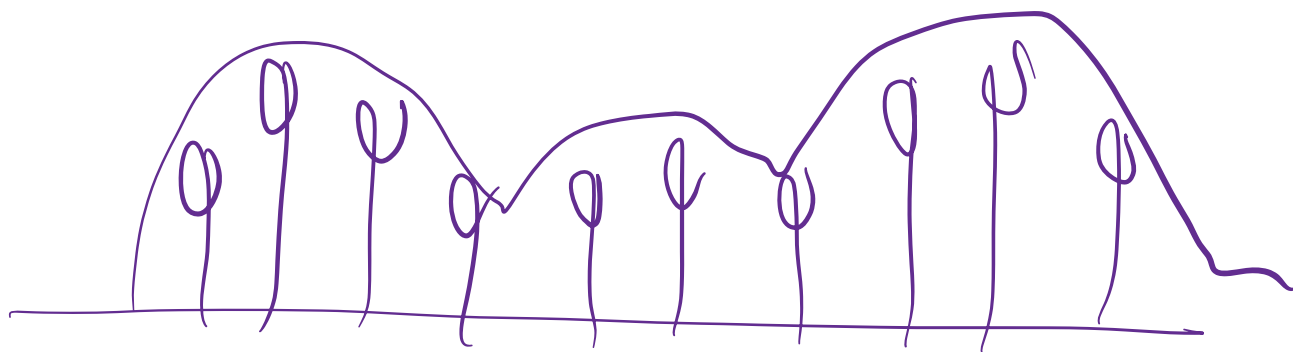
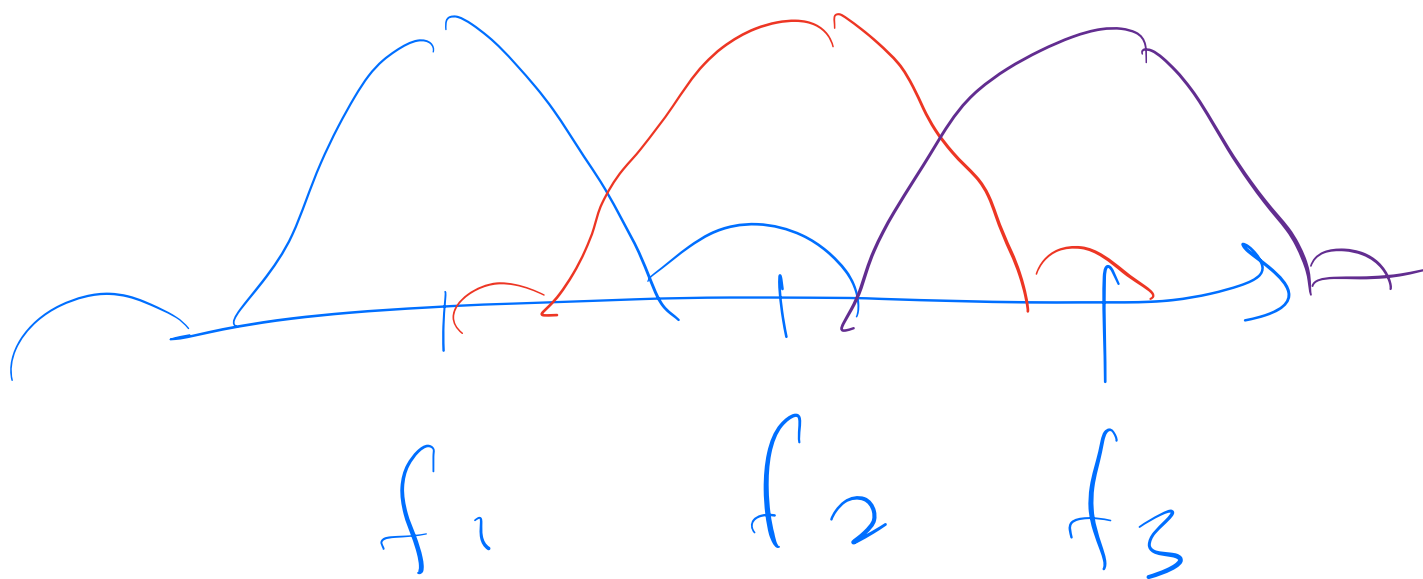
multiply by
rect
window

in
Time domain



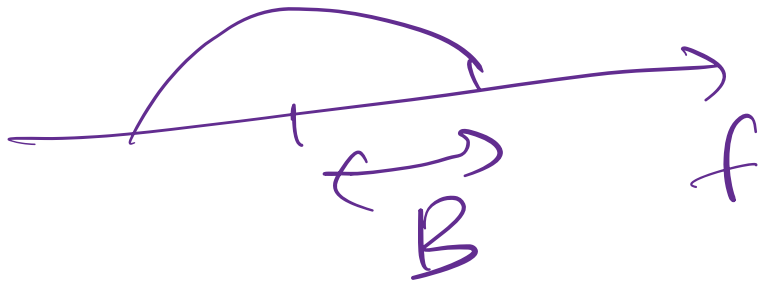
convolve
with
sinc





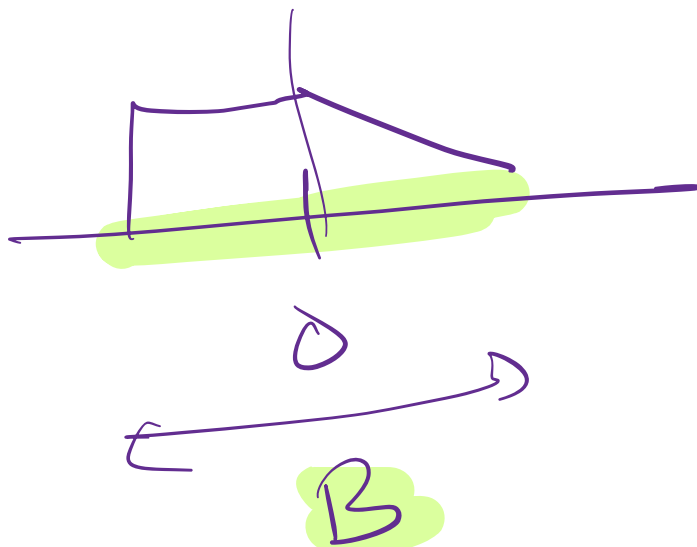
Nyquist criteria

(1) real signal

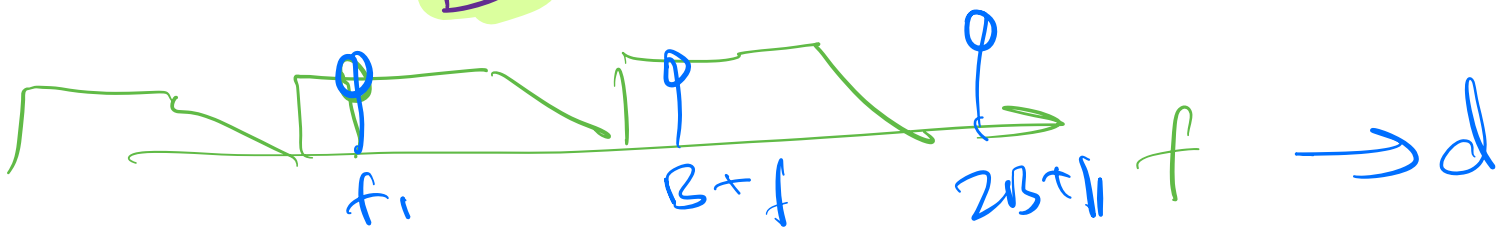


→ nyquist rate
 $f_{\text{nyq}} > 2B$

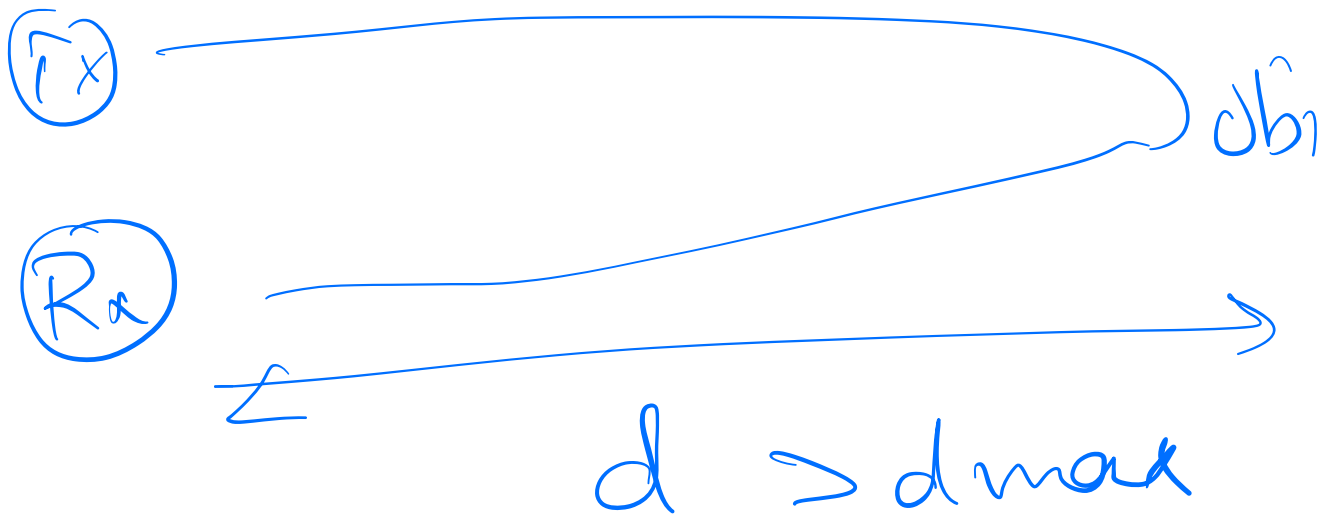
(2) complex signal



→ nyquist
 $f_{\text{nyq}} > \underline{B}$

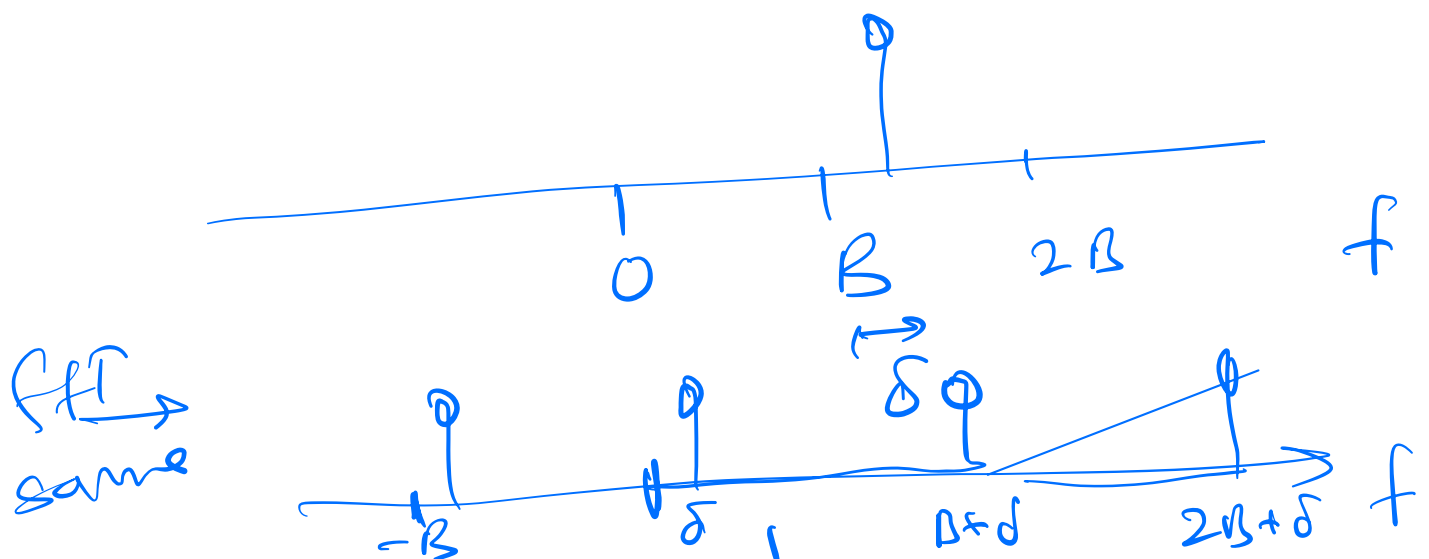


what happen if an
object is very far
 $d > d_{max}$



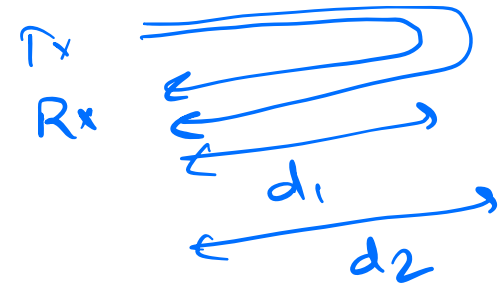
$$\Rightarrow \Delta f > B \quad \left| \quad \begin{array}{l} \text{Case 2} \\ \Delta f = \delta \end{array} \right.$$

$$\Delta f = B + \delta$$



FM CW Range Resolution

$\min(d_2 - d_1)$
can be resolved



- FM CW Transmitted Signal:

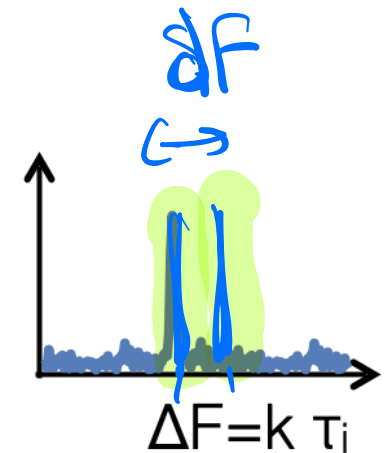
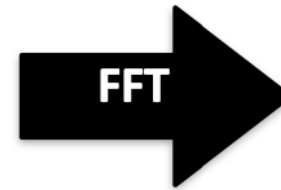
$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

- FM CW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FM CW after downconversion:

$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$



- Sampling Rate = B

$$\Delta F < B \longrightarrow \tau_{\max} = B/k = B \times T_s / B_s \longrightarrow d_{\max} = c \times B \times T_s / 2 B_s$$

- Sampling Window = T_s

$$\boxed{\Delta F > 1/T_s} \longrightarrow \tau_{\min} = 1/(k \times T_s) = 1/B_s \longrightarrow \boxed{d_{\min} = c/2B_s}$$

$$z_{\min} = \frac{dF}{k}$$

$$B_s = k T_s$$

$$d_{\min} = z_{\min} \times \frac{c}{2}$$

given range resolvable 2 cm
& max range 10 m

given $B = 100 \text{ KSPS}$

what is B_s ? $\int K = \frac{B_s}{T_s}$
what is T_s ?

$$d_{\min} = \frac{c}{2B_s} \Rightarrow B_s = \frac{2}{c} d_{\min}$$

$$d_{\max} = \frac{c}{2} \frac{B T_s}{B_s}$$

$$T_s = \frac{2 d_{\max}}{c} \frac{B_s}{B}$$

$$\Rightarrow K = \frac{B_s}{T_s}$$

FMCW Range Estimation Summary

$$y_b(t) = \sum_i \underbrace{(A_i e^{j f_0 \tau_i})}_{\text{constant for path } i} \underbrace{e^{j 2\pi k \tau_i t}}_{\text{time var.}}$$

- FMCW Transmitted Signal:

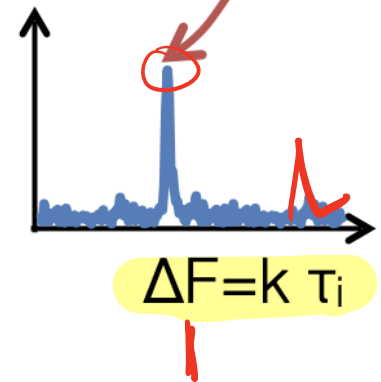
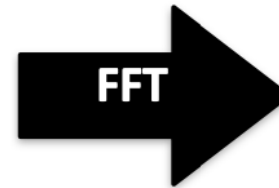
$$x(t) = e^{j 2\pi (\frac{k}{2} t^2 + f_0 t)}$$

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j 2\pi (\frac{k}{2} (t - \tau_i)^2 + f_0 (t - \tau_i))}$$

- FMCW after downconversion:

$$y_b(t) = \sum_i A_i e^{j 2\pi (k \tau_i t + f_0 \tau_i)}$$



for each path → can identify magnitude & phase

- Phase of peak = $f_0 \tau_i$

- Phase wraps around 2π

- Use peak position $\Delta F = k \tau_i$ for course estimate of τ_i

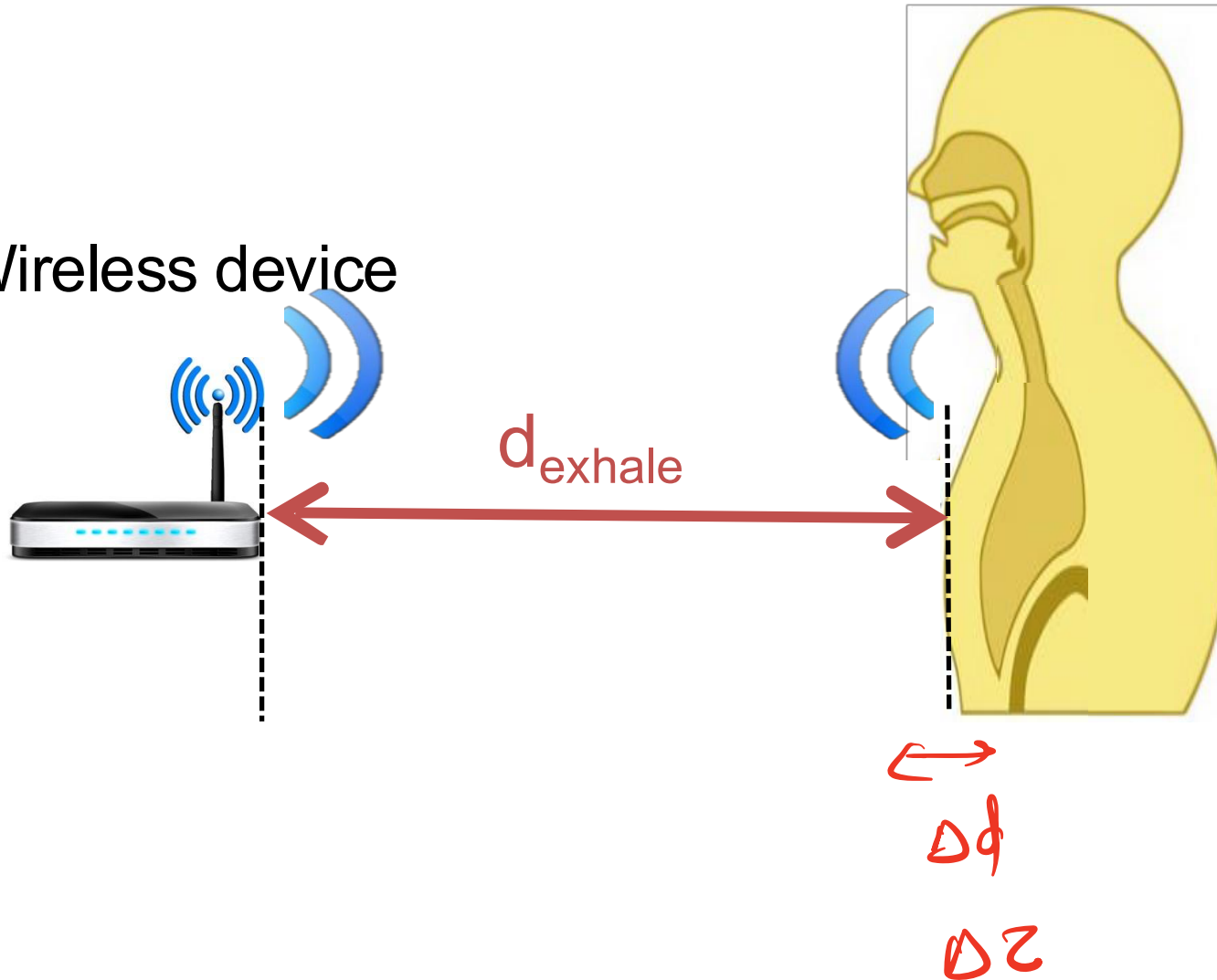
- Use peak phase $f_0 \tau_i$ for fine estimate of τ_i

to center freq

Vital Radio: Use wireless reflections off the human body to monitor breathing and heart rate

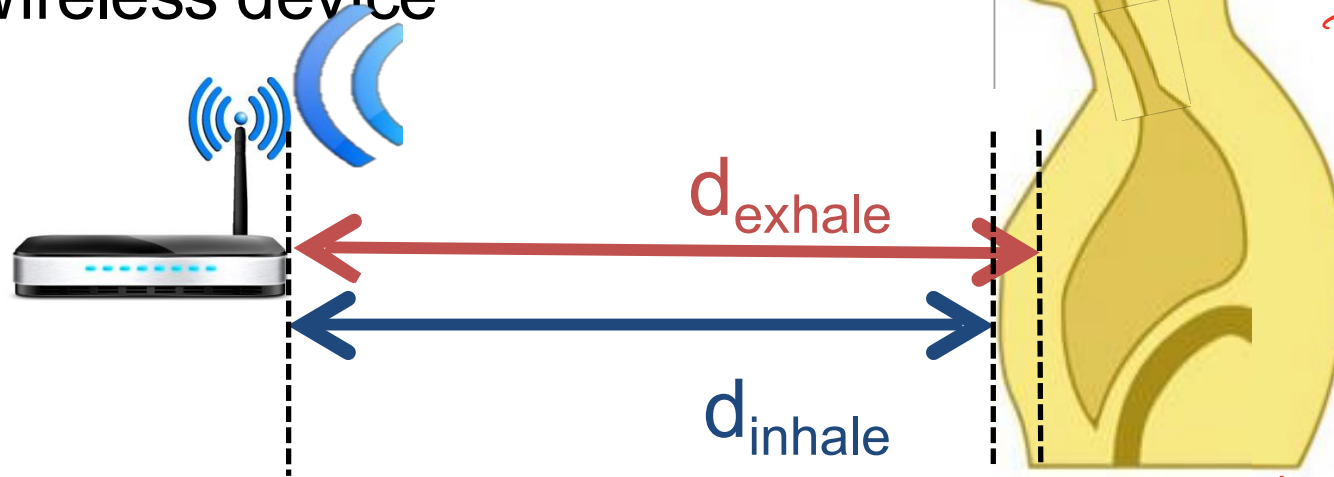
Mobicom'14: <https://witrack.csail.mit.edu/vitalradio/>

Wireless device



Problem: Localization accuracy is only 12cm and cannot capture vital signs

Wireless device



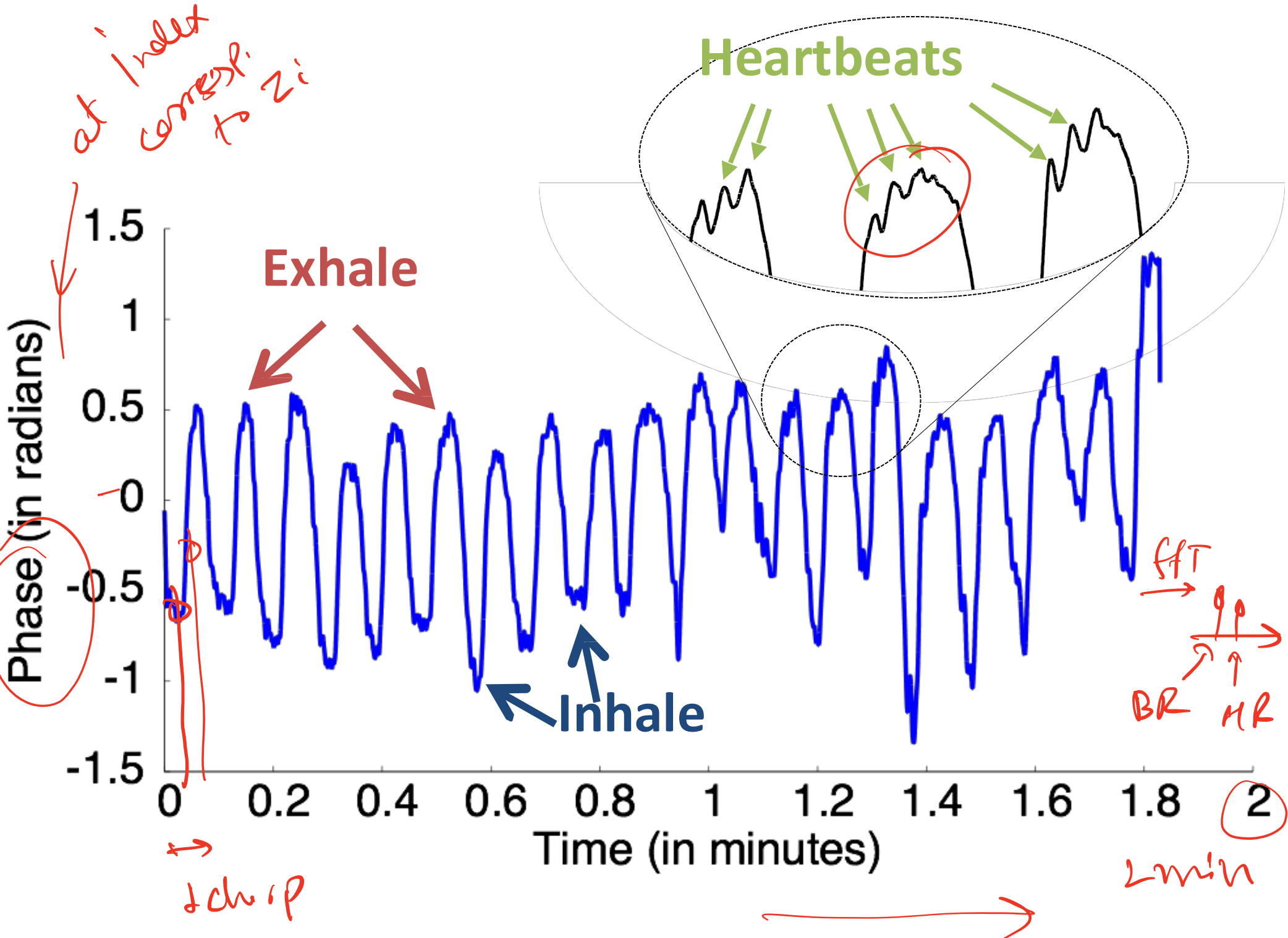
resolution
 $z_0 \approx \frac{1}{B_s}$
 $d \approx \frac{c}{2} \frac{1}{B_s}$

$d_{\text{exh}} - d_{\text{inh}} < c \cdot d_{\text{res}}$

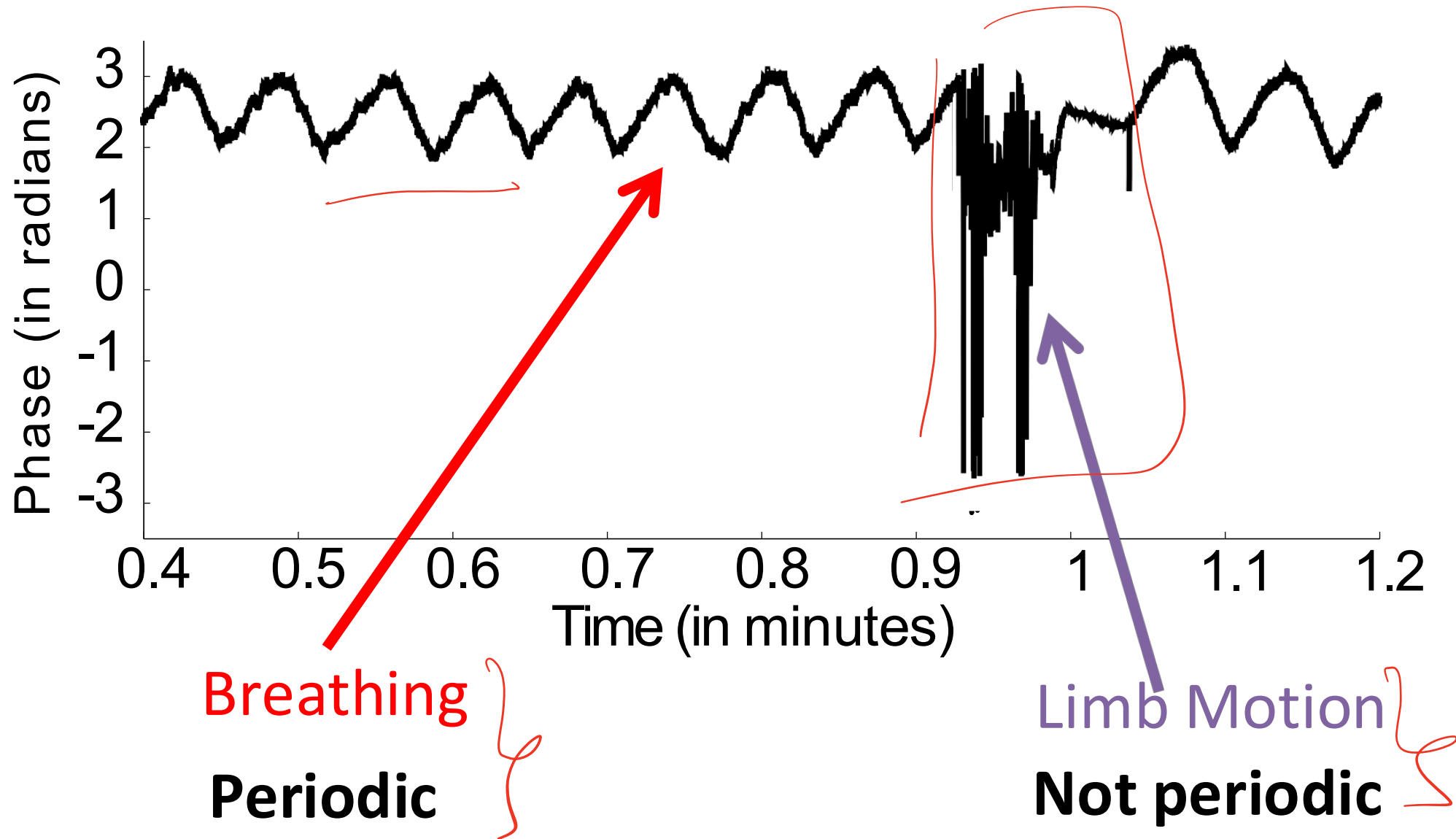
phase at this index

z_i

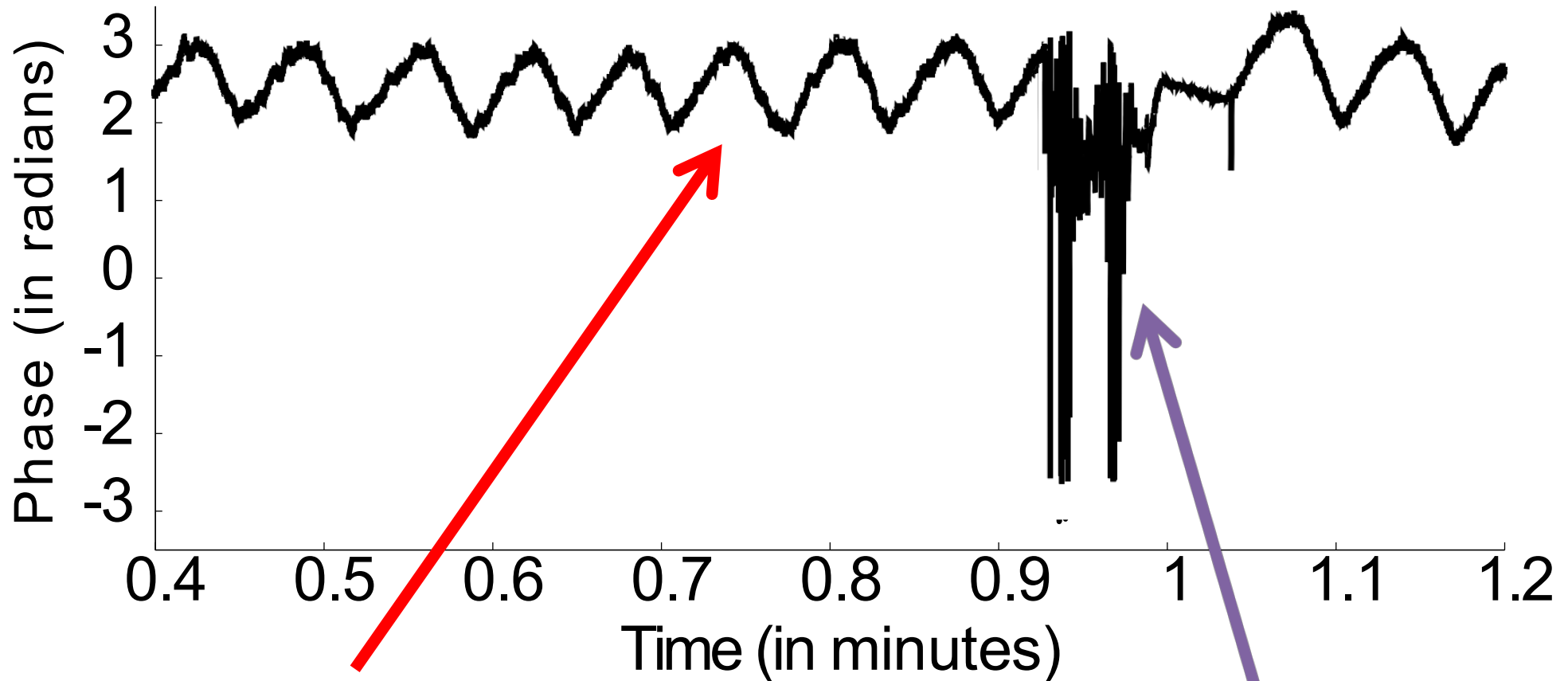
f/z



What happens when a person moves his limb?

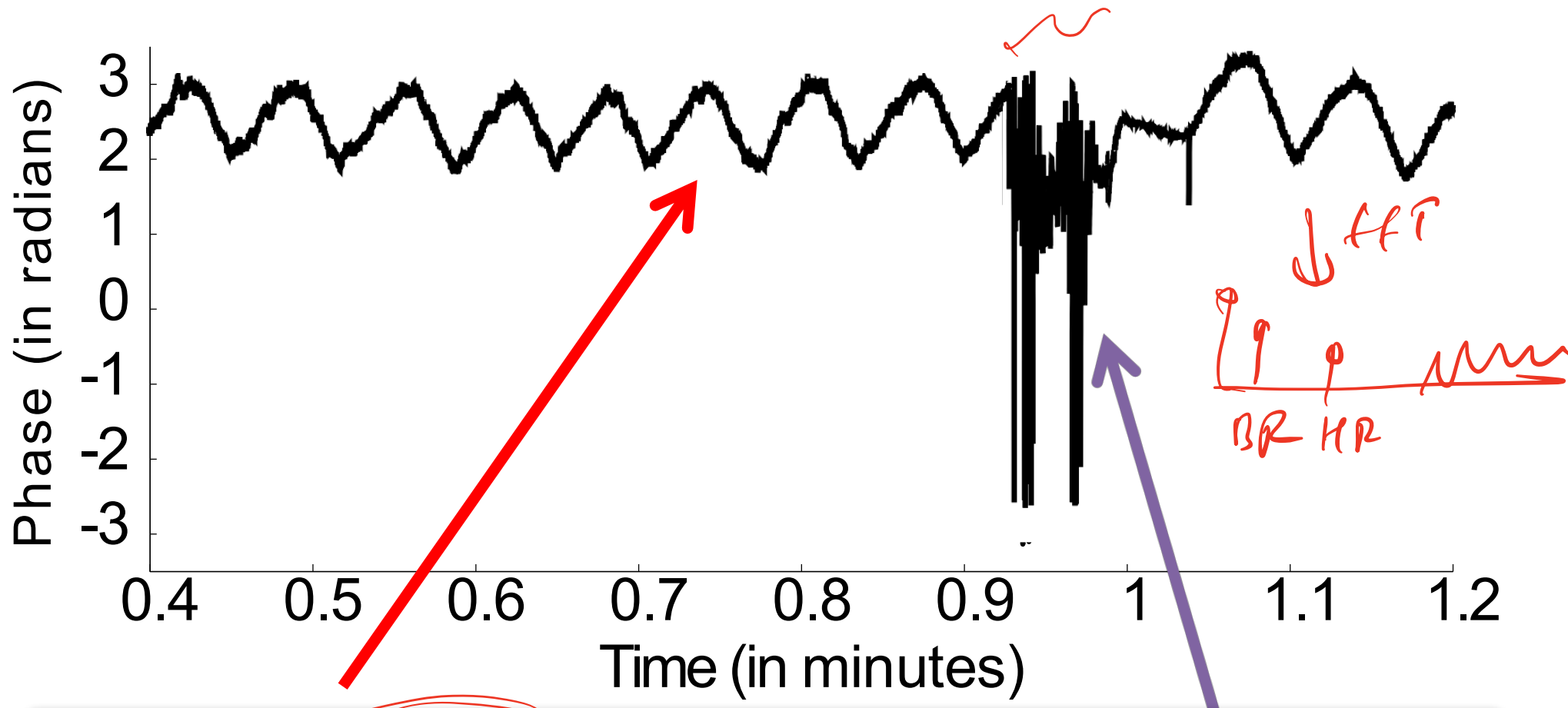


What happens when a person moves his limb?



Use periodicity test to eliminate variations that are not due to breathing/heartbeats

What happens when a person moves his limb?

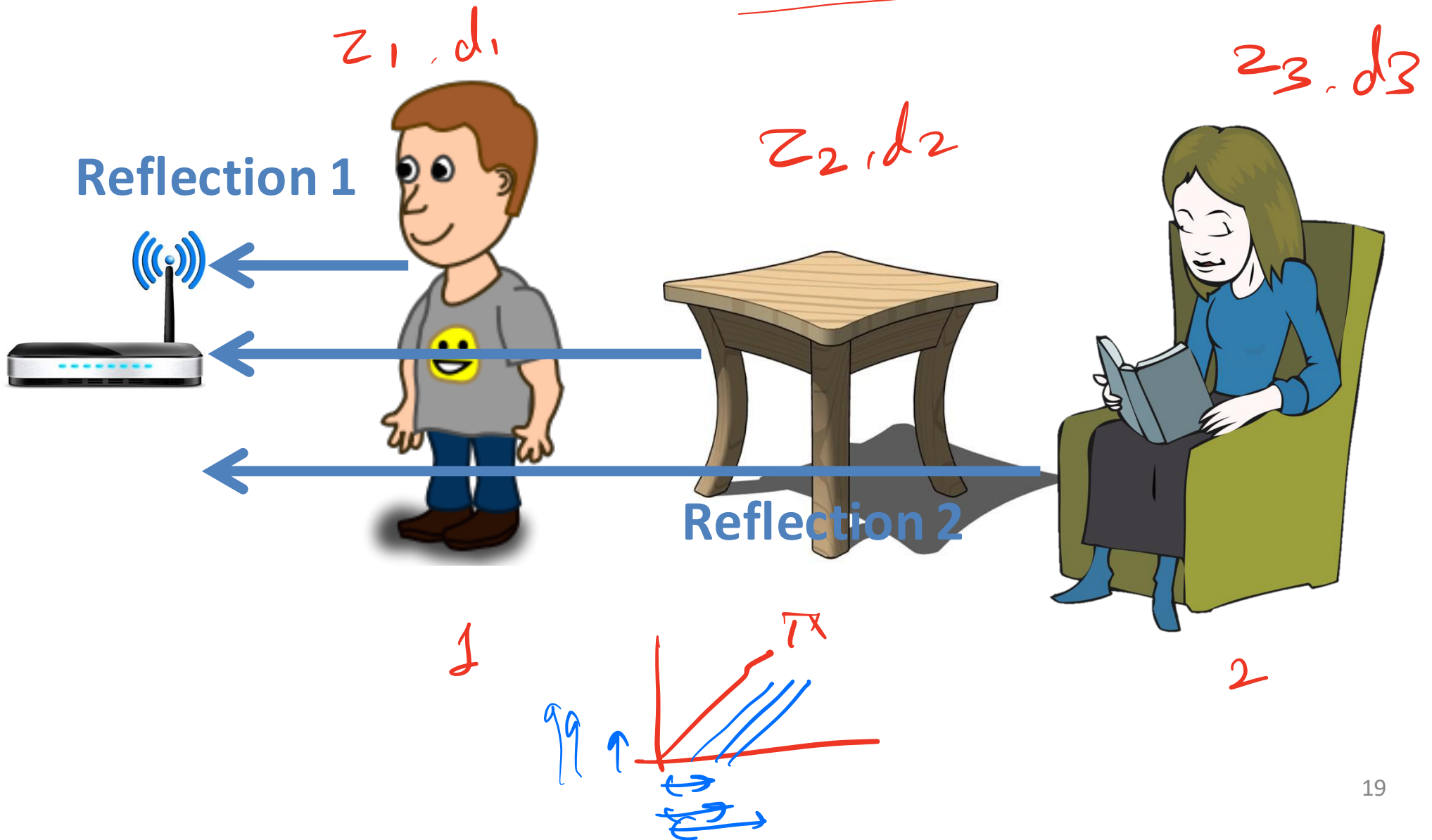


Band-pass filter the cleaned signals to extract breathing and heart rate

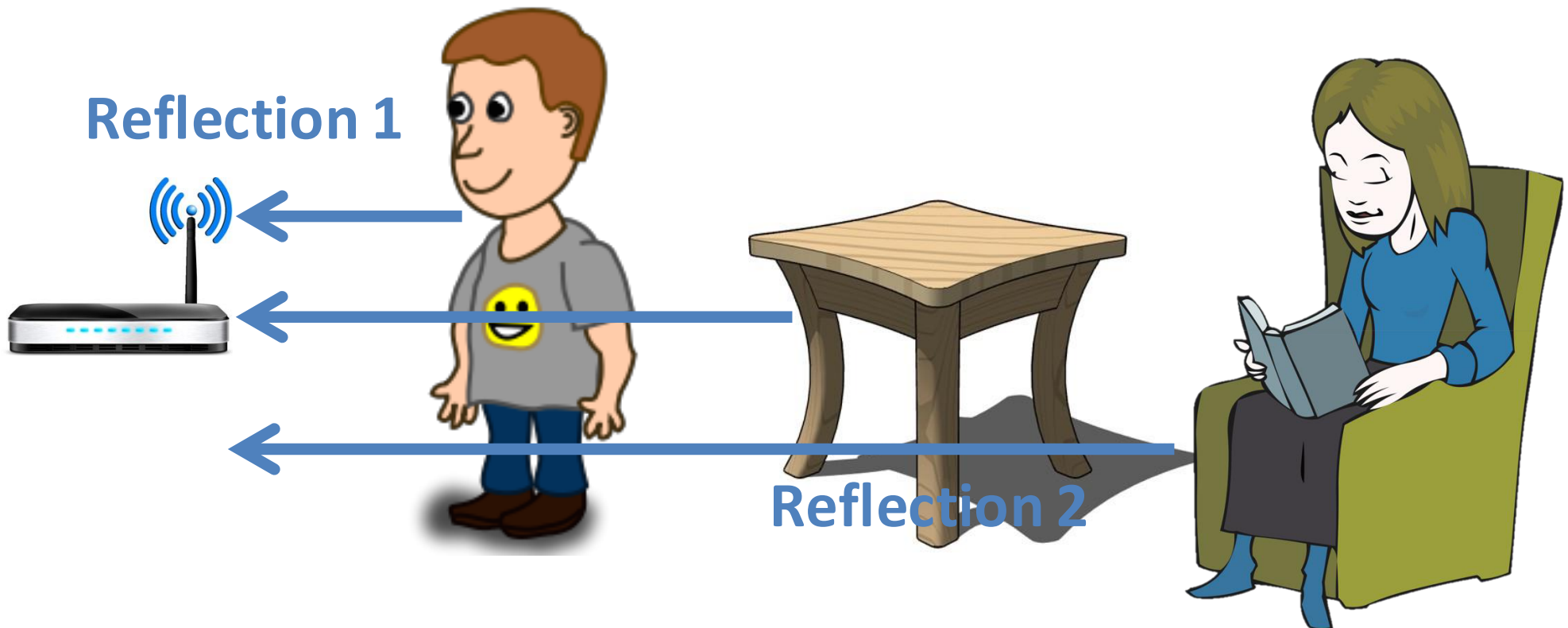
What happens with multiple users in the environment?

Reflections from different objects **collide**

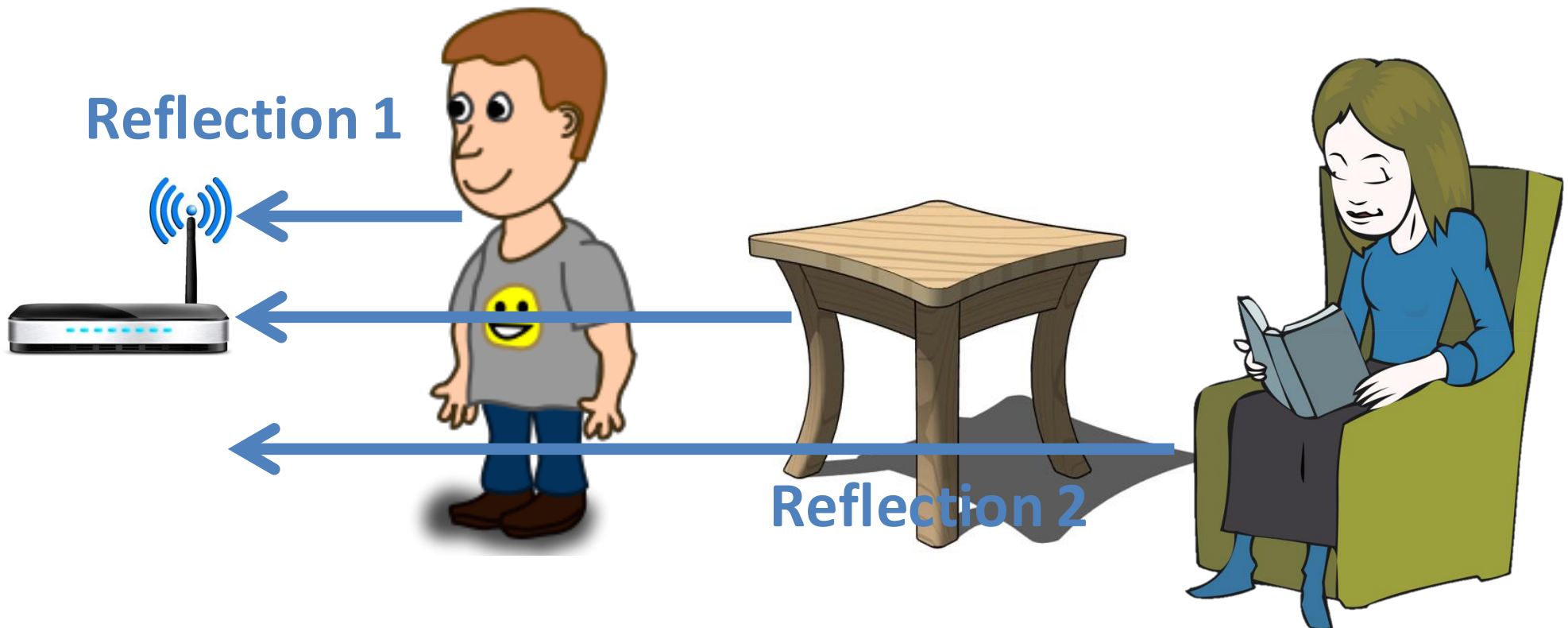
Problem: Rx receives sum of multipaths



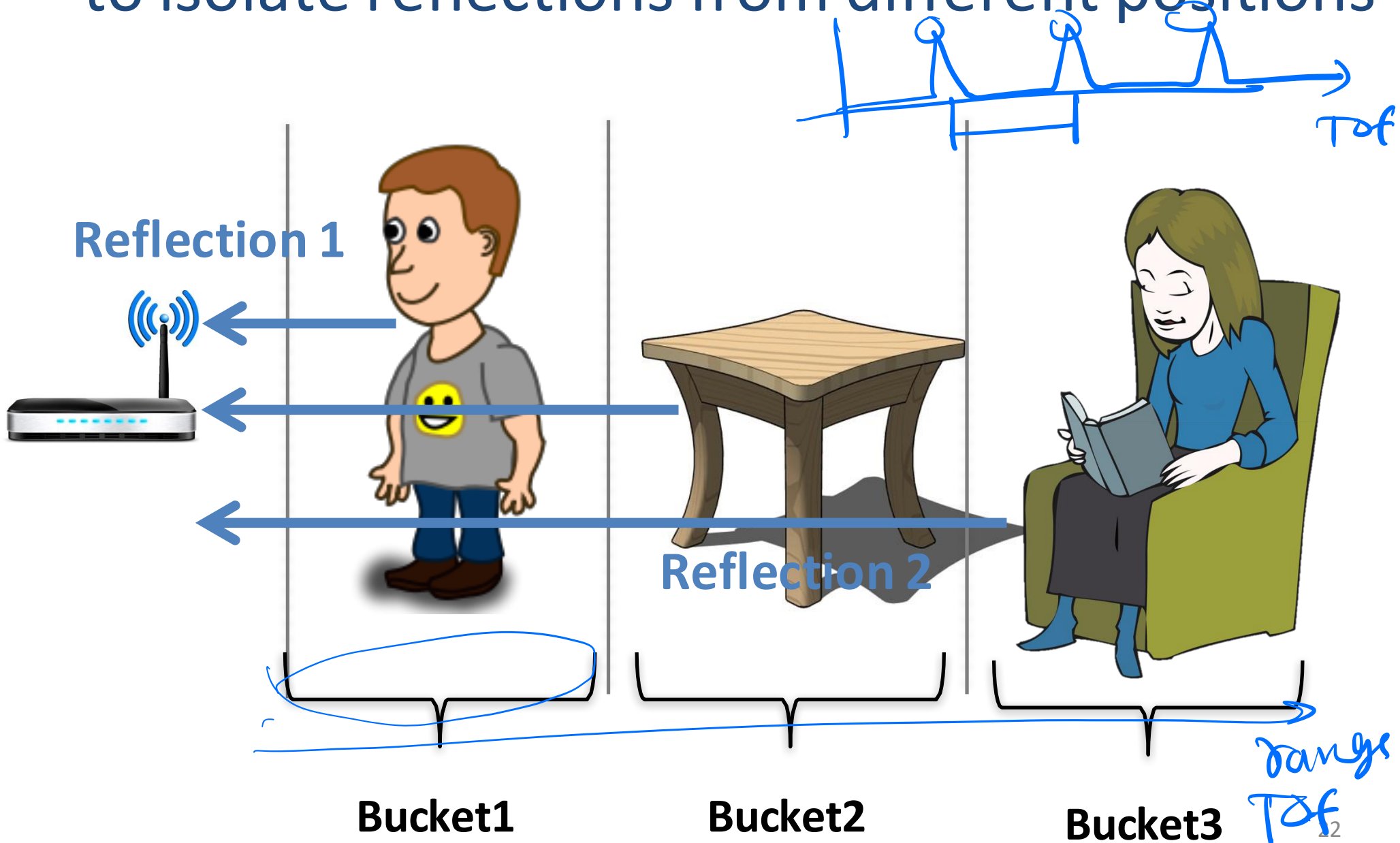
Idea: **Wireless localization** can be used to locate various devices



Solution: Use **wireless localization as a filter** to isolate reflections from different positions



Solution: Use **wireless localization as a filter** to isolate reflections from different positions



Summary: Multi-person Breathing

Step 1: Transmit a wireless signal and capture its reflection

Step 2: Isolate reflections from different objects based on their positions

Step 3: Zoom in on each object's reflection to obtain phase variations due to vital signs

Step 4: Use frequency analysis to separate breathing and heart rate signals

Vital-Radio Limitations

- Minimum separation between users: 1-2m

range resolution!

- Monitoring range: 8m

⇒ limited

⇒ lim. sample rate

sweep Bandwidth

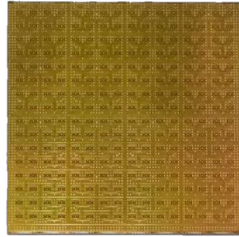
- Collects measurements when users are quasi-static

Can we use millimeter wave radars for 3D imaging and not just ranging?

5G pushing research into delivering large 2D millimeter wave phased arrays



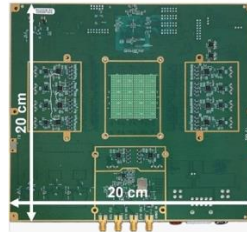
Nokia & National Instruments



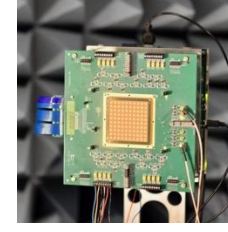
UCSD
256 elements



UCSD
64 elements



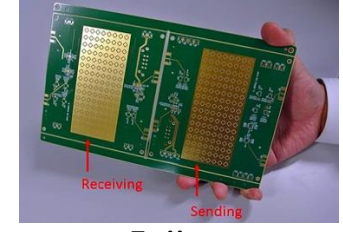
Bell Labs
384 elements



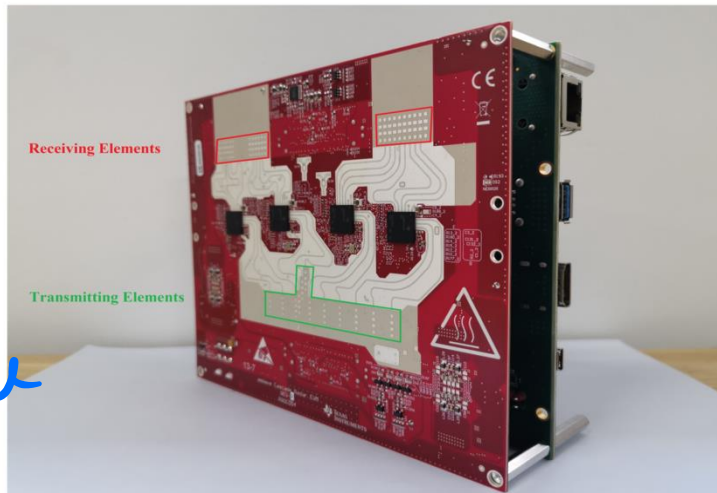
Anokiwave
256 elements



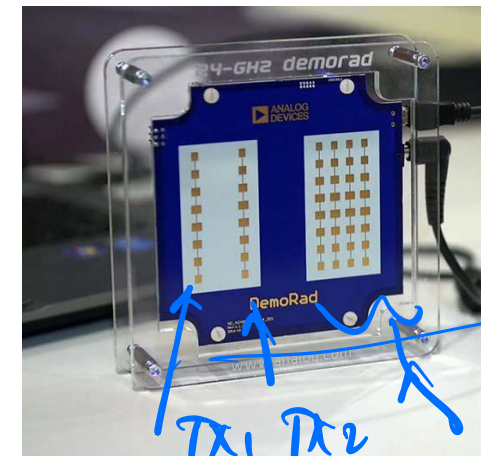
IBM
64 elements



Fujitsu
64 elements



TI Cascade Radar



Analog Devices TinyRad

Small wavelength enables thousands of antennas to be packed into small space

➔ Extremely narrow beams

range
angle
y
n

Rx1
Rx2
Rx3
Rx4

Challenges in mmWave Imaging

1. Extremely Low Resolution

- Blobs of radar reflections
- No Sharp Boundaries/Shapes

2. Specularity

- Major parts of Car are Missing

3. Multipath and artifacts

- Spurious Reflections

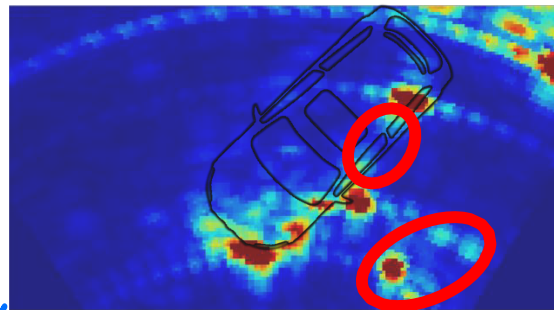
(a) Camera Image



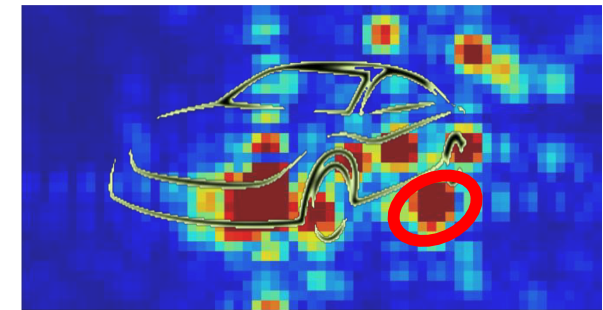
(b) Radar Point Cloud



(c) Top-View of Radar Heatmap



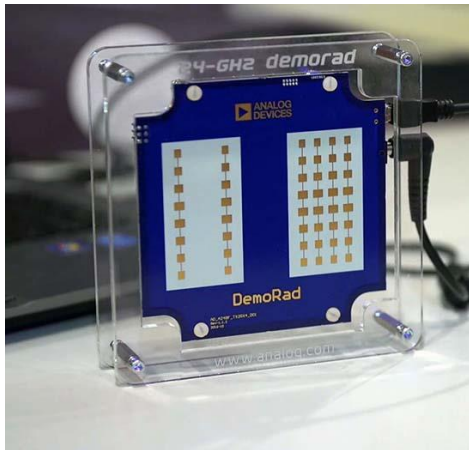
(d) Front-View of Radar Heatmap



Radar images lack perceptual and contextual information about the scene

RF Imaging

$\text{phase}(f, z_i)$ for each antenna

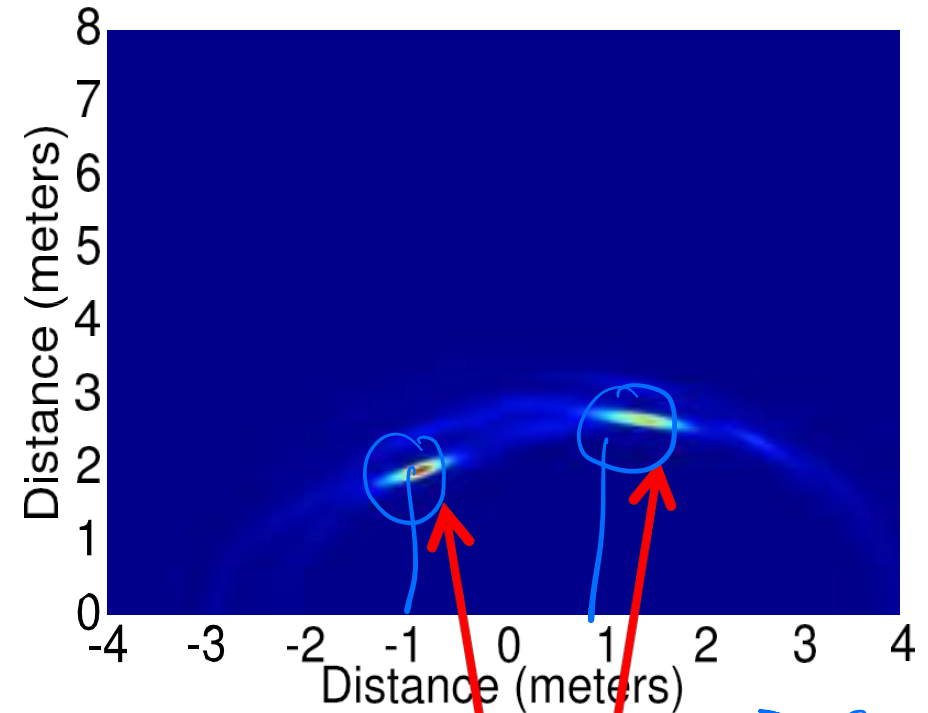


Range-Angle FFT



Transform to x-y coordinate system

People are points



Localize the two users

$(f, z_i)_1$ $(f, z_i)_2$

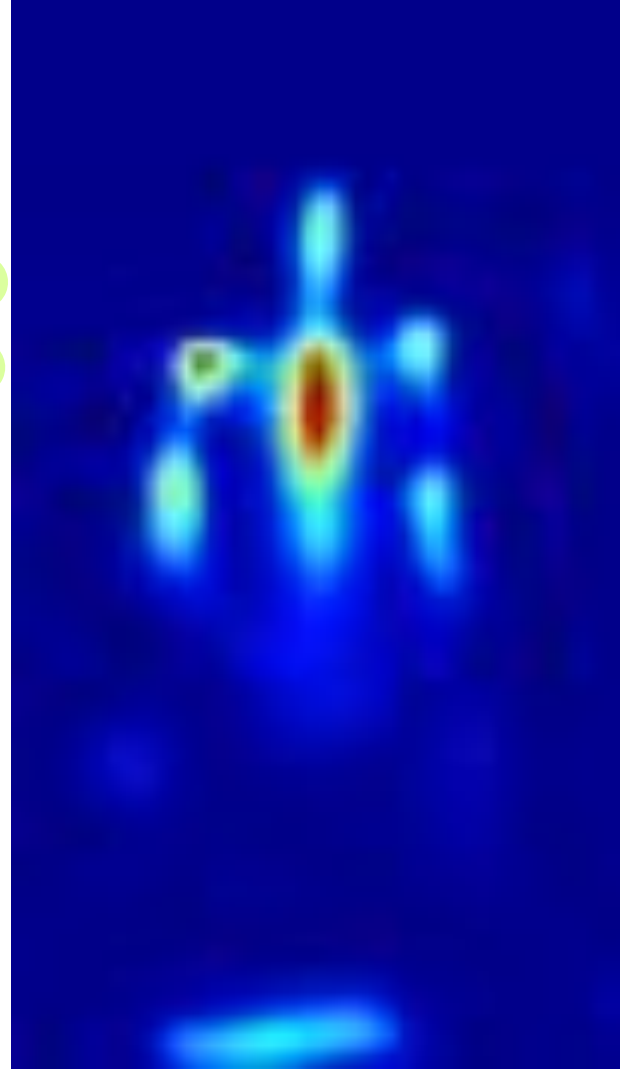
$z_{i1}, z_{i2} \dots$

$\frac{d \sin \theta}{c}$

$\frac{2d \sin \theta}{c}$

$\frac{2d \sin \theta}{c}$

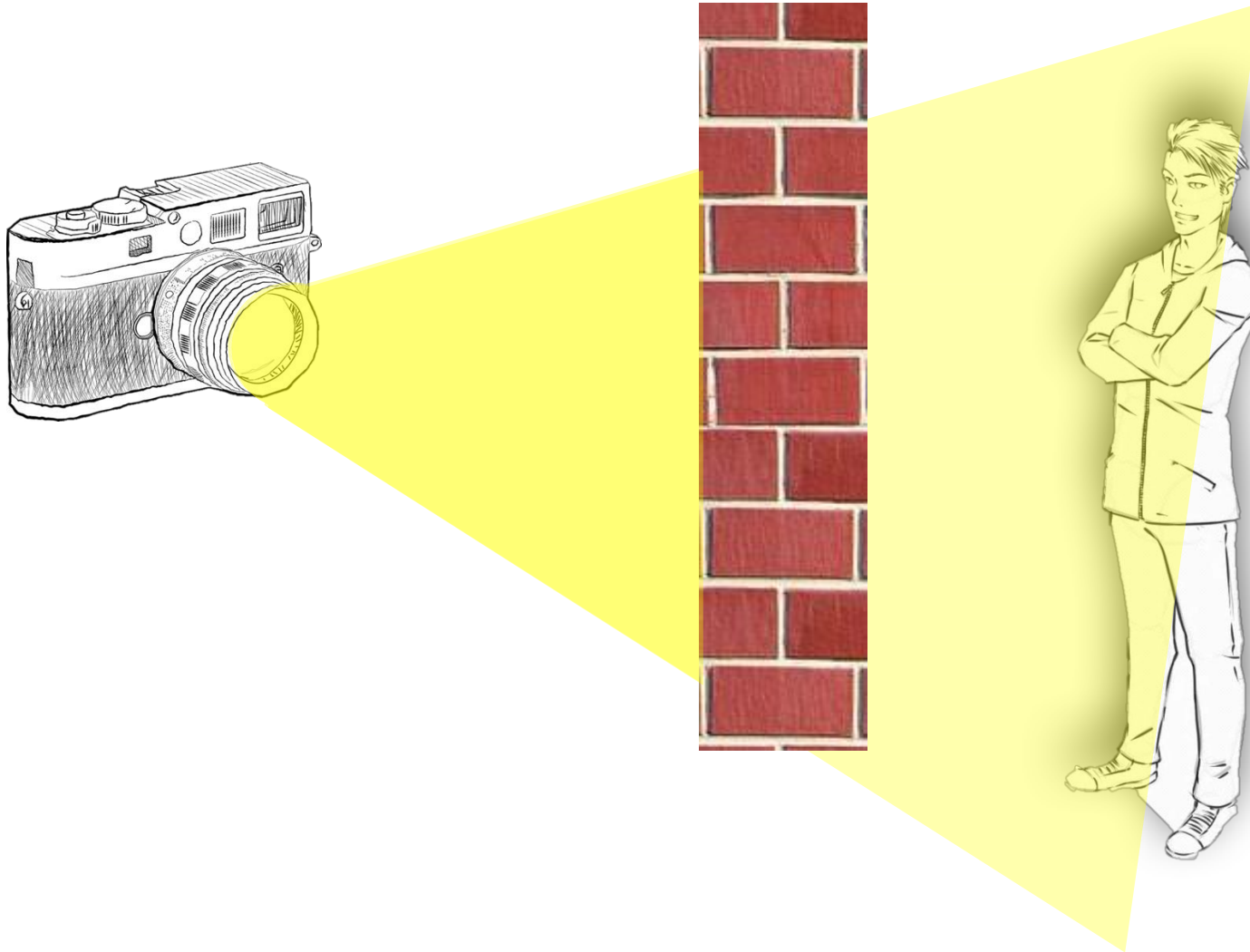
Capturing a Coarse Human Silhouette



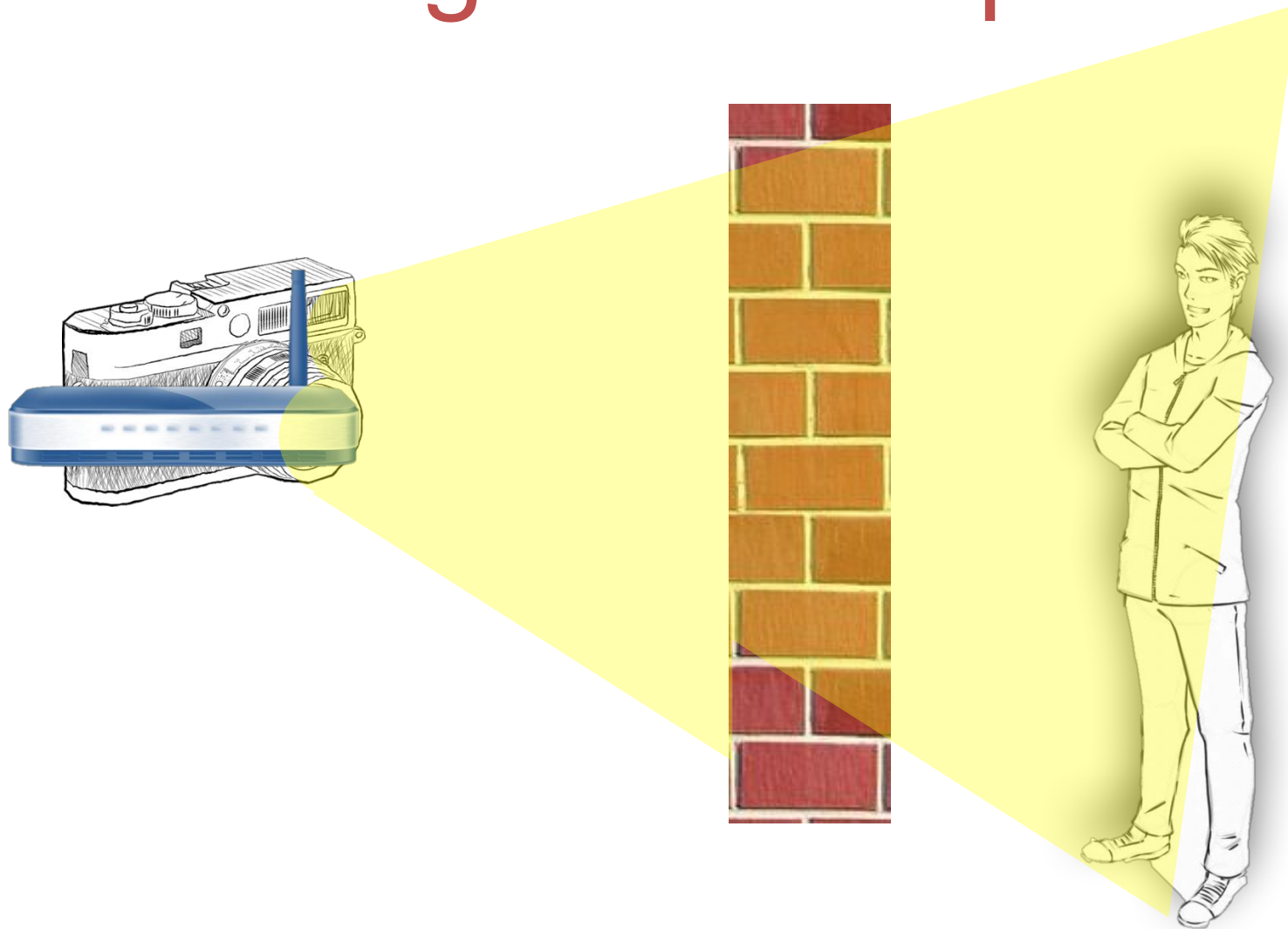
RFCapture: [https://dspace.mit.edu/handle/111588](https://dspace.mit.edu/handle/1721.1/111588)

RFPose: <https://rfpose.csail.mit.edu/>

Imaging through occlusions



Imaging through occlusions using radio frequencies



Traditional Imaging

Cannot image through occlusions like walls

Form 2D images using lenses

Get a reflection from all points: can image all the body

RF Imaging

 Walls are transparent and can image through them

 No lenses at these frequencies

 No reflections from most points: all reflections are specular

RF Imaging



Walls are transparent and can image through them



No lenses at these frequencies



Our Solution: A component that scans 3D space with RF and outputs reflection snapshots at every point in time



No reflections from most points: all reflections are specular

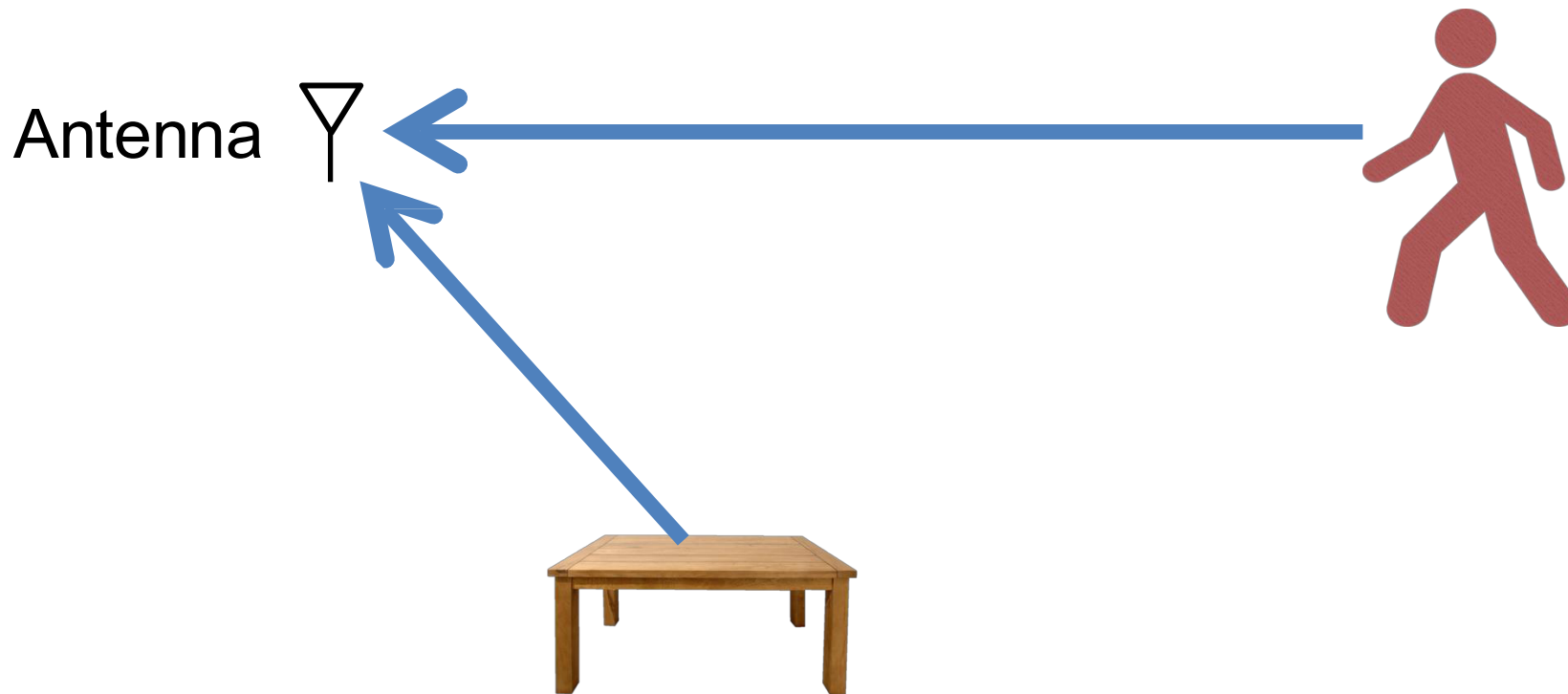


Advanced DSP and ML Techniques

Imaging with RF

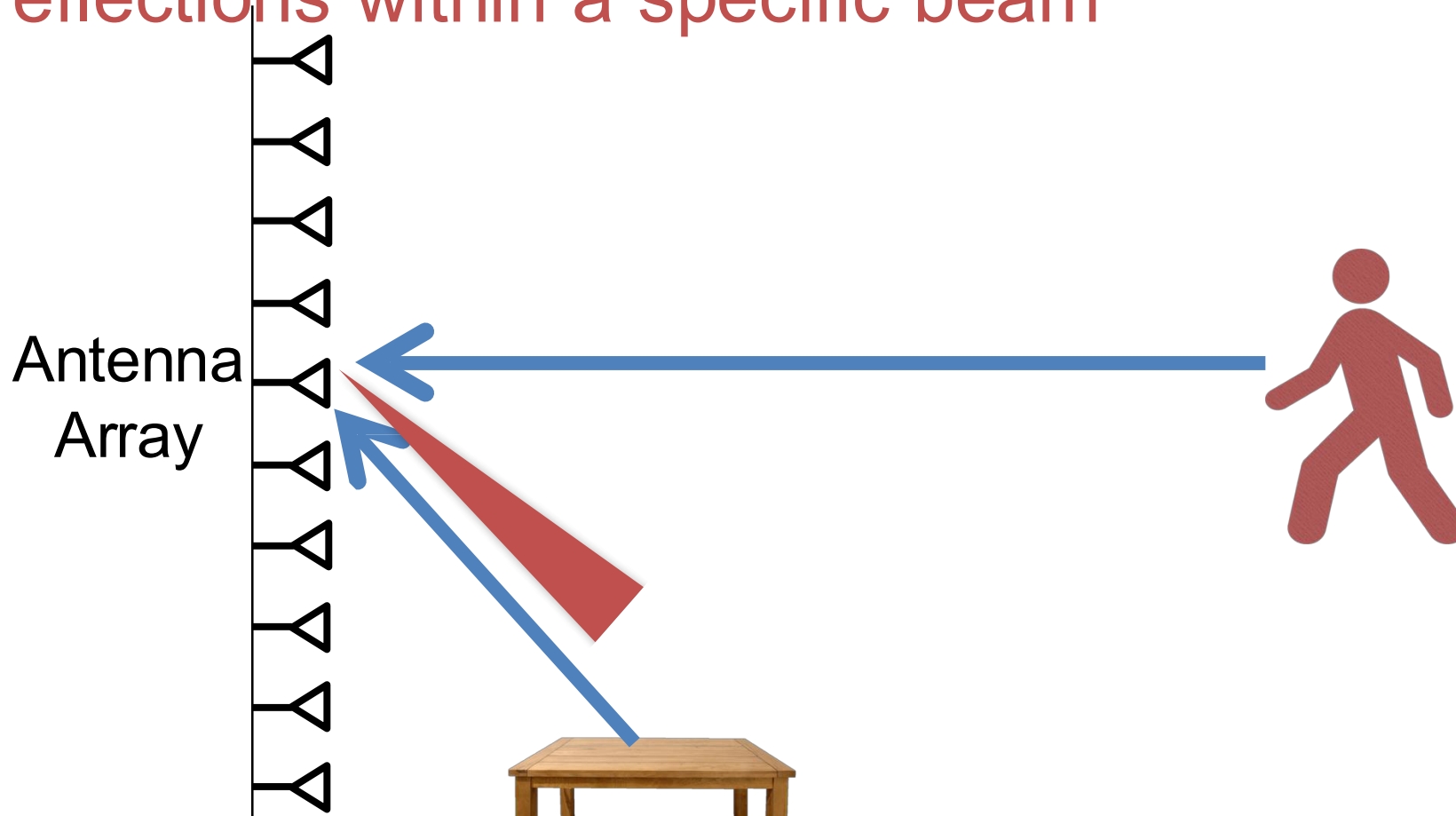
No lens at these frequencies

Antenna cannot distinguish bounces from different directions



Imaging with RF

Beamforming: Use multiple antennas to scan reflections within a specific beam



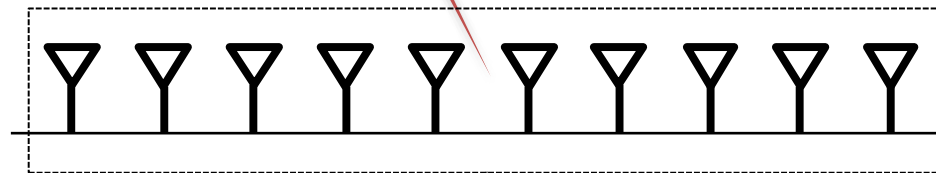
Extend to 3D with time-of-flight measurements by repeating this at every depth

Scanning every direction is slow

- Each angle/depth needs to be processed separately
- Most of the 3D scene is empty
- Our solution: Coarse-to-fine scan that iteratively refines the resolution

Coarse-to-fine Scan

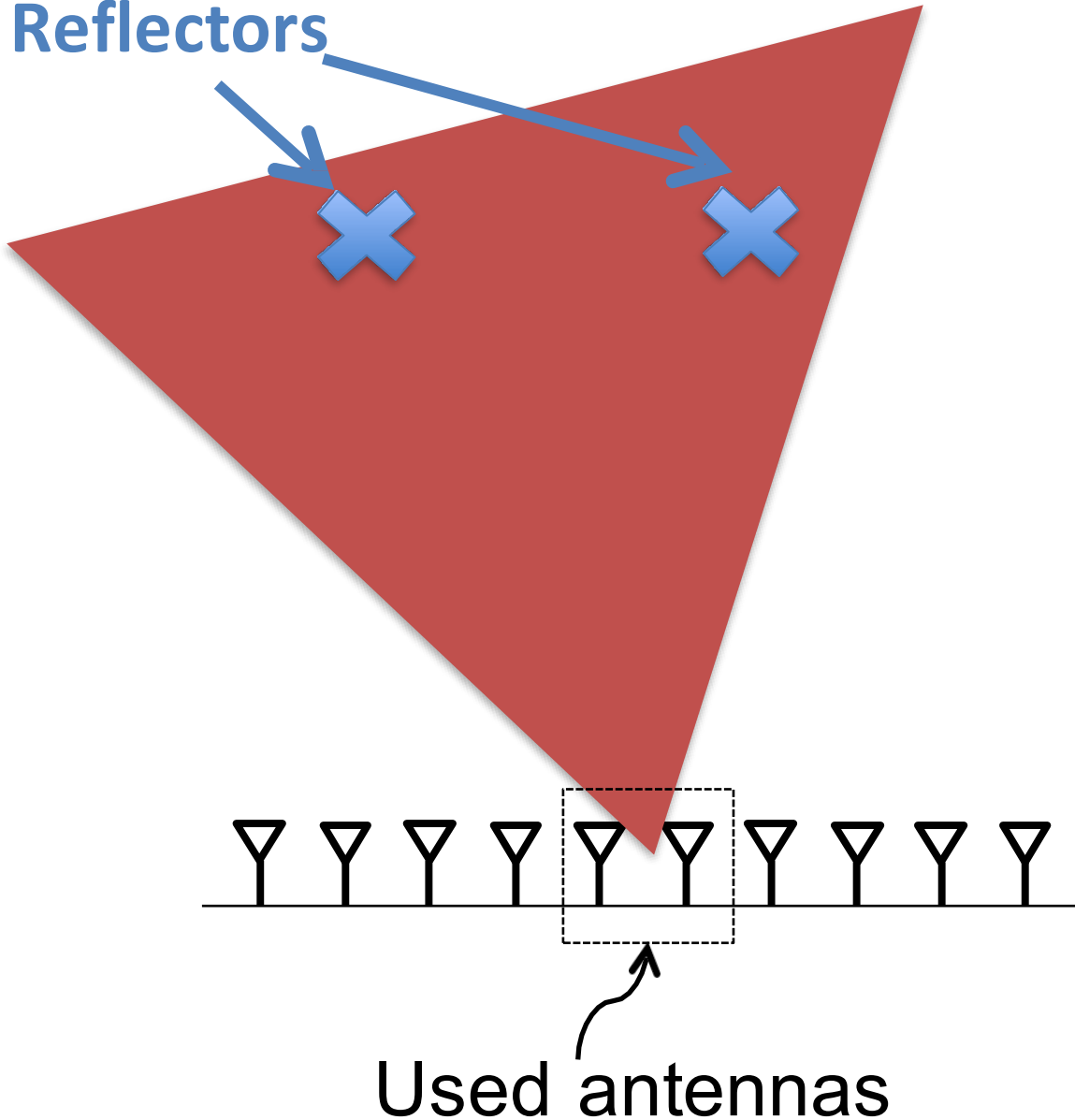
- Larger aperture (more antennas) means finer resolution



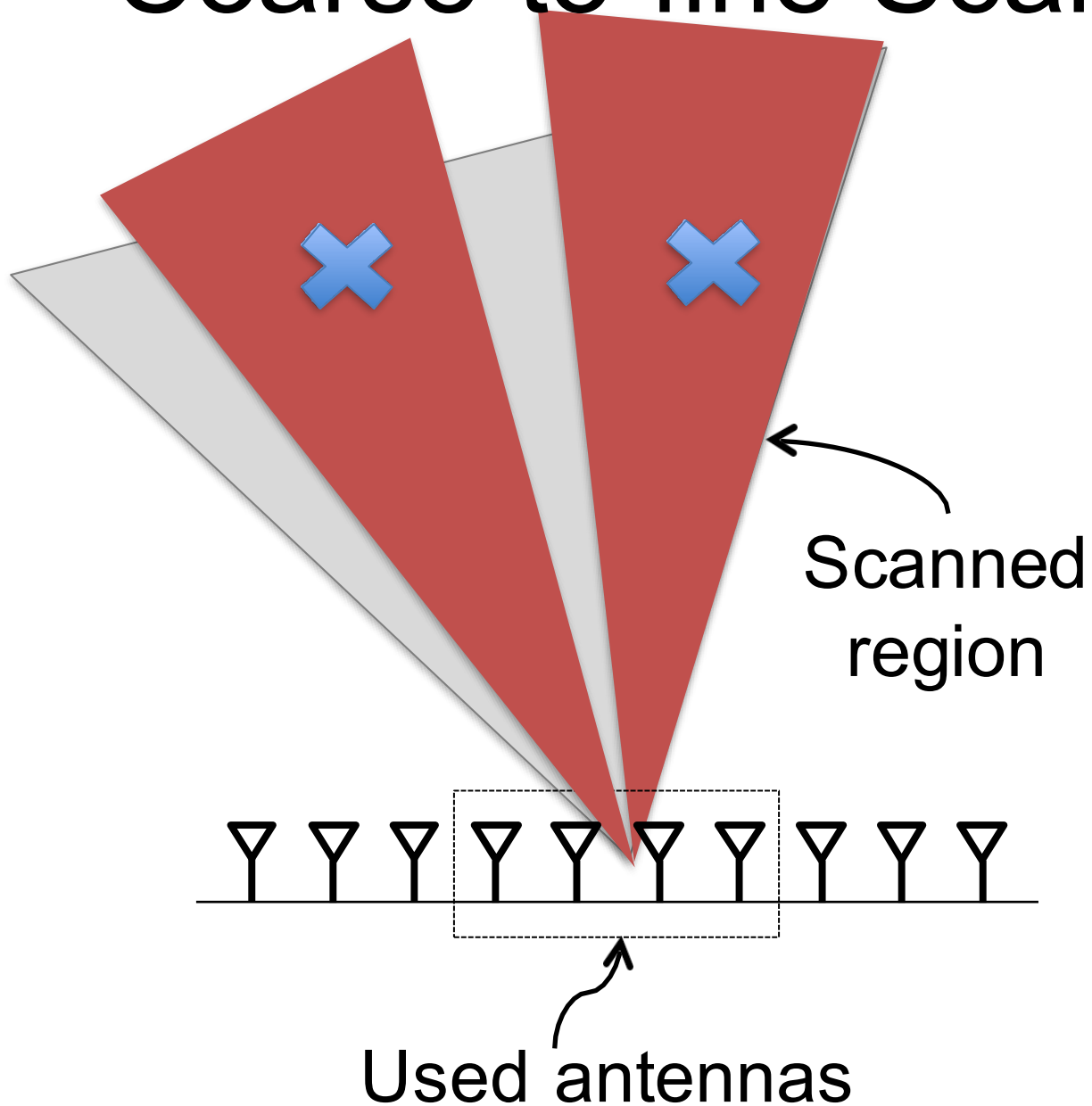
Used antennas

Coarse-to-fine Scan

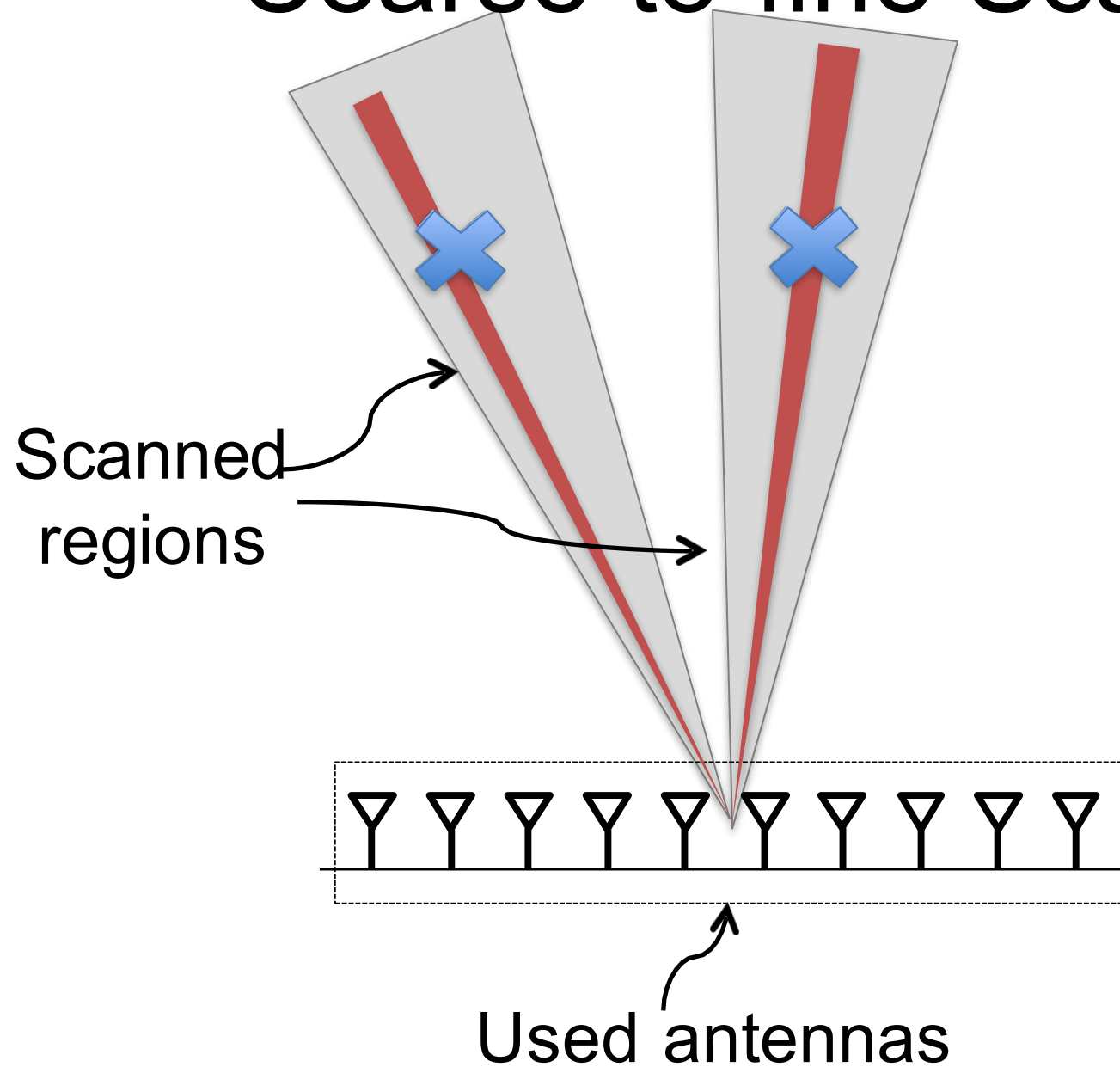
Reflectors



Coarse-to-fine Scan



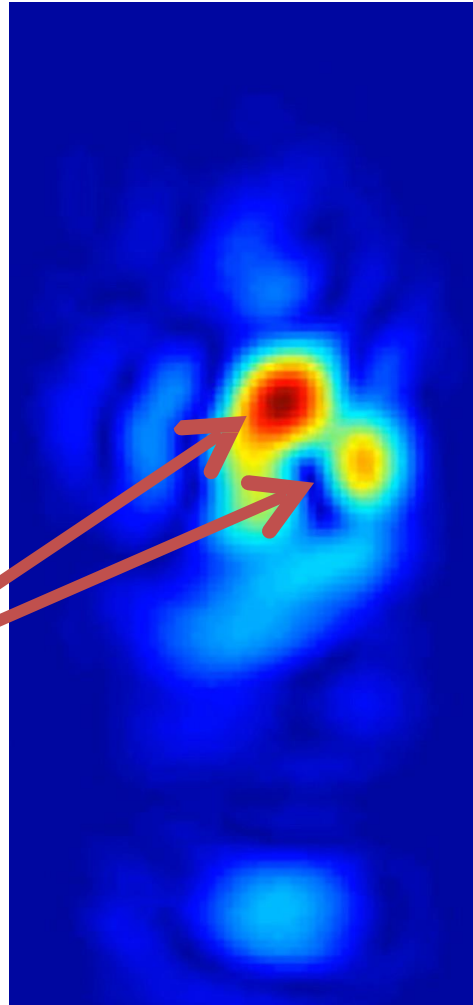
Coarse-to-fine Scan



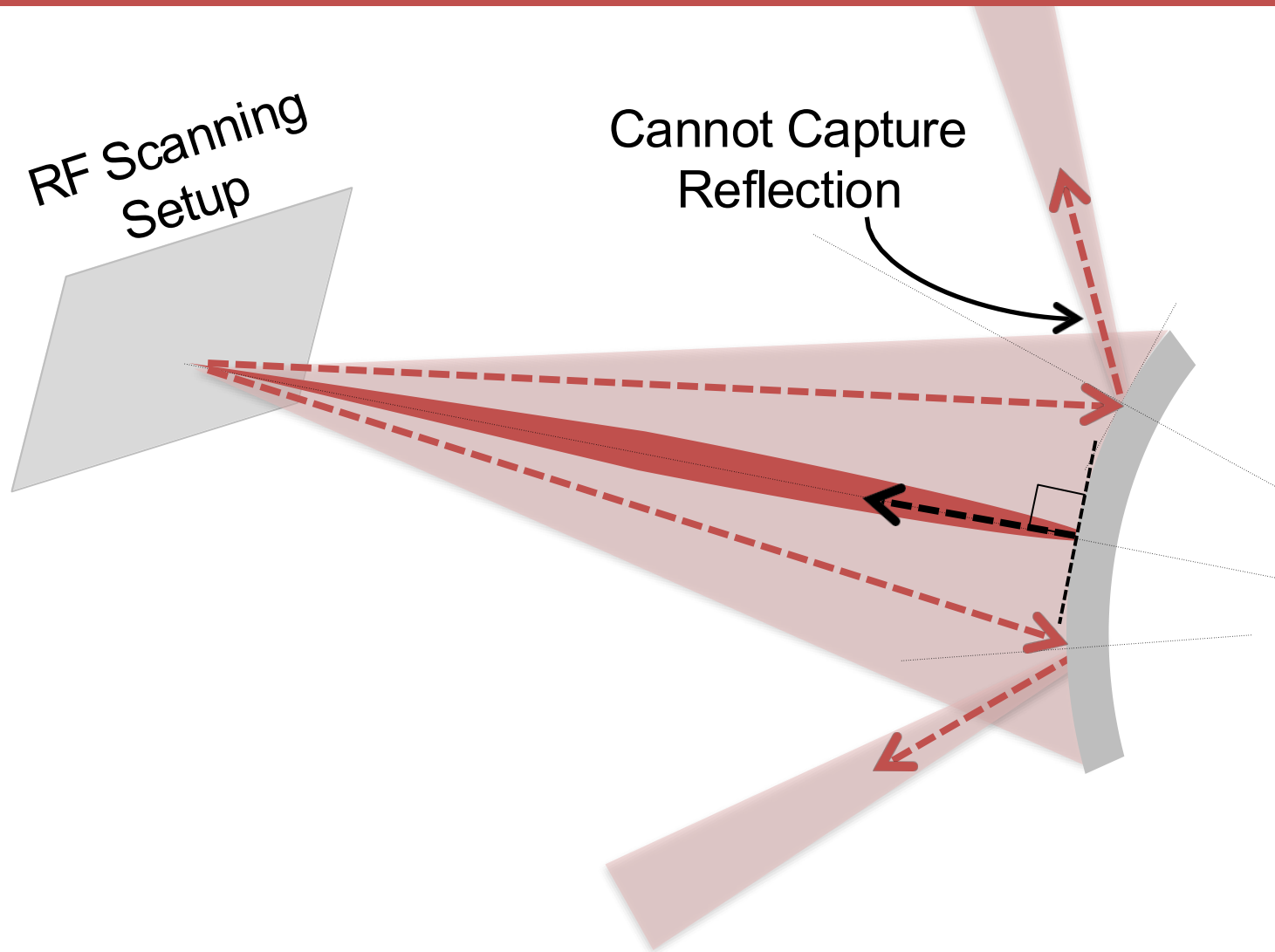
Challenge: We only obtain blobs in space

Output of 3D RF Scan

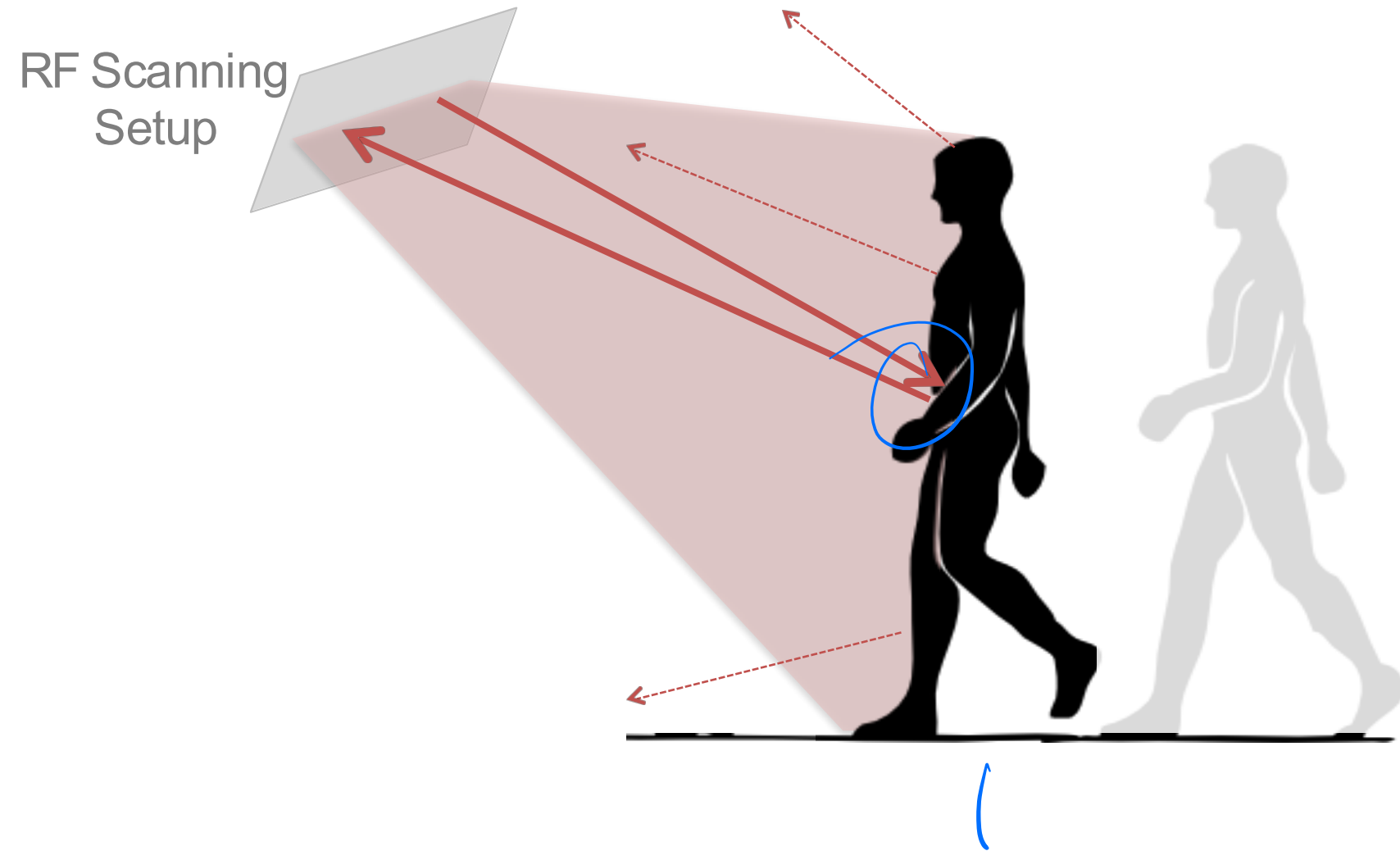
Blobs of
reflection power



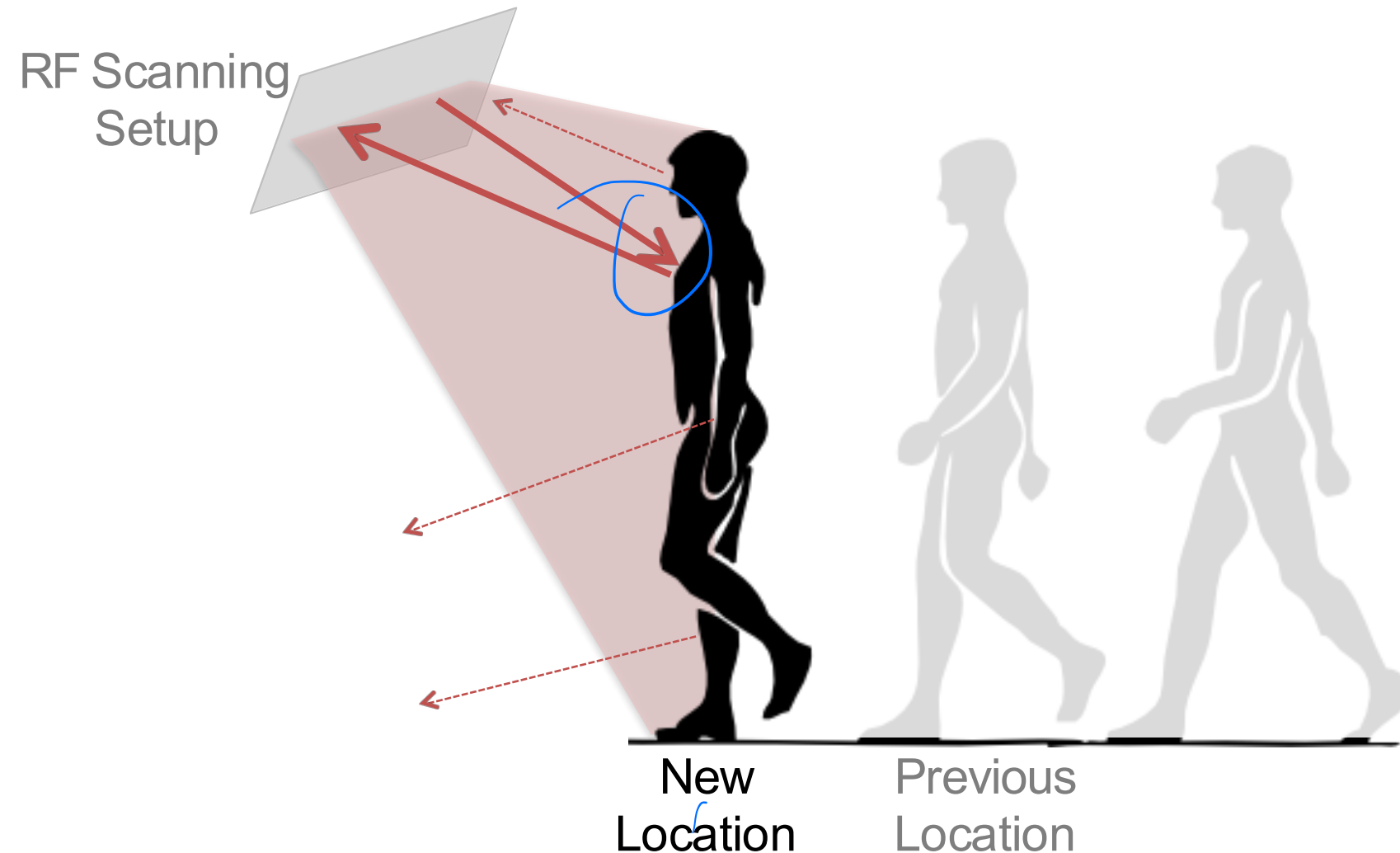
At every point in time, we get reflections from only a subset of body parts.



Solution Idea: Exploit Human Motion and Aggregate over Time



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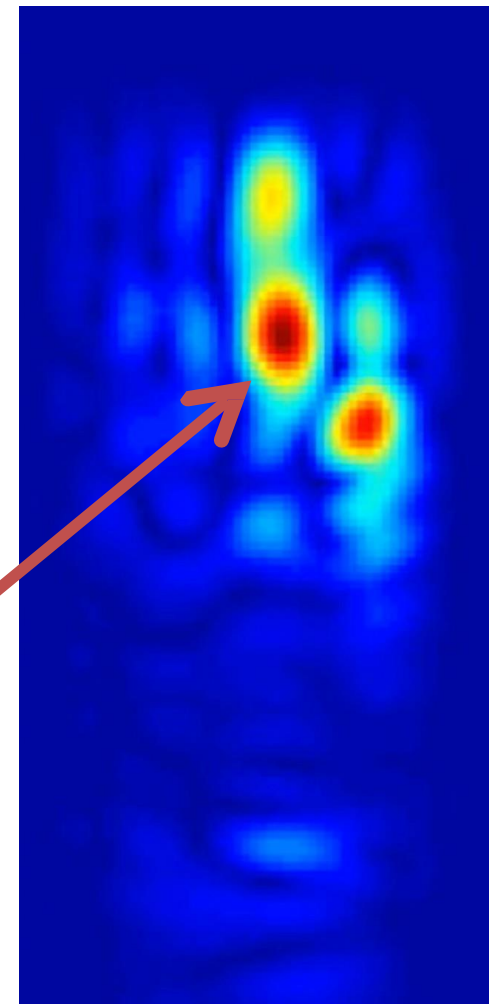
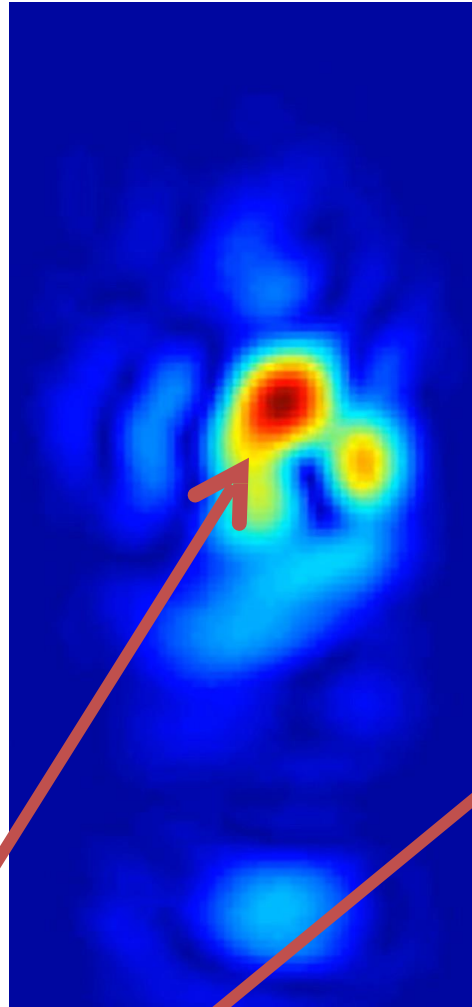
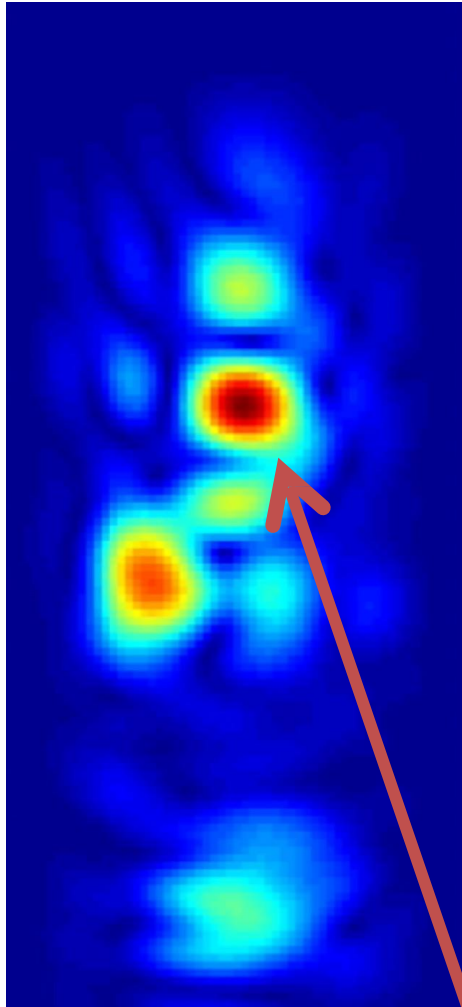
Combine the various snapshots

Human Walks toward Sensor

3m

2.5m

2m



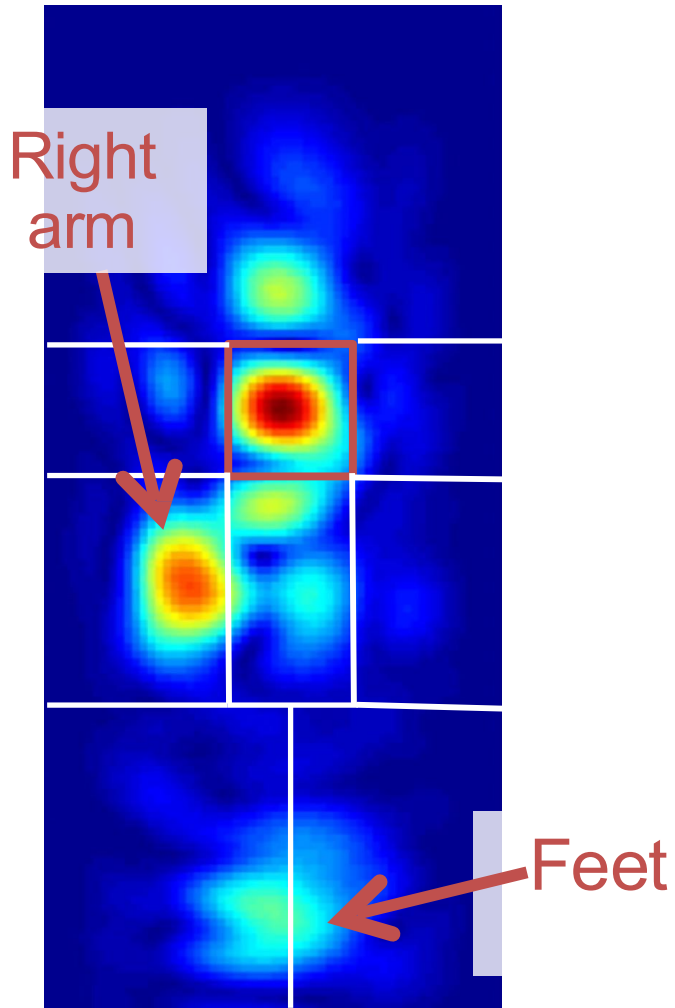
Chest (Largest
Convex Reflector)



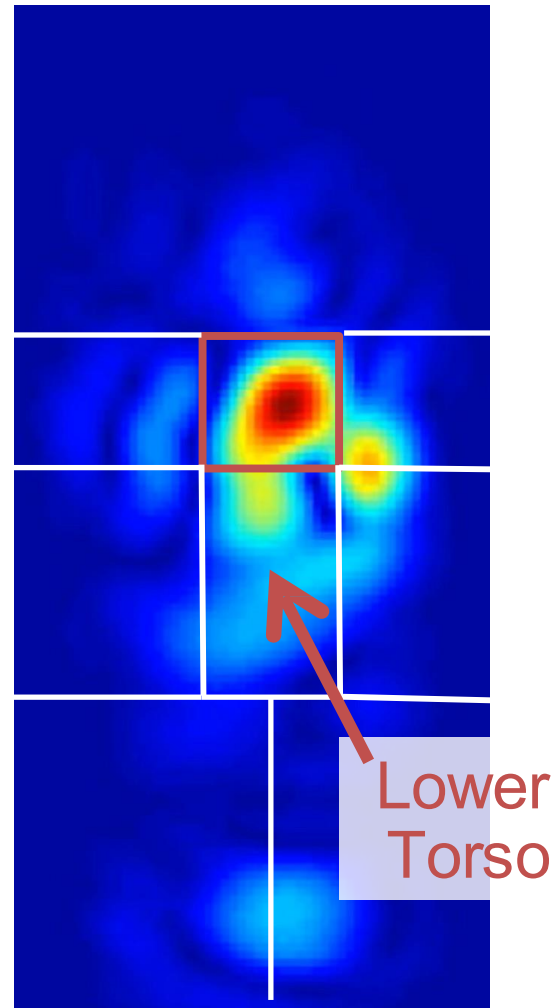
Use it as a pivot: for motion
compensation and segmentation

Human Walks toward Sensor

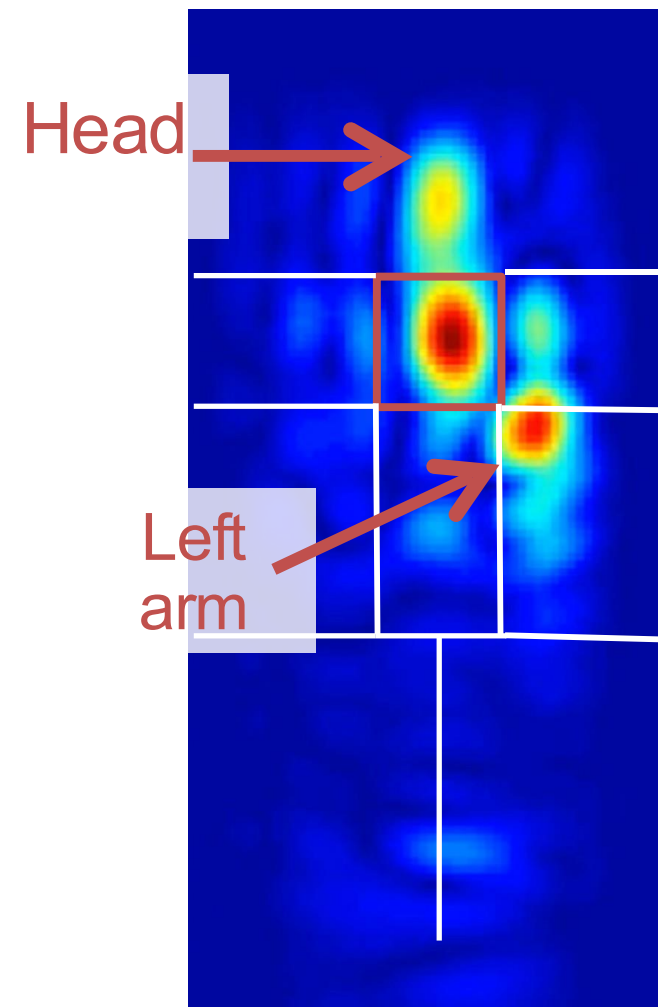
3m



2.5m



2m



Combine the various snapshots

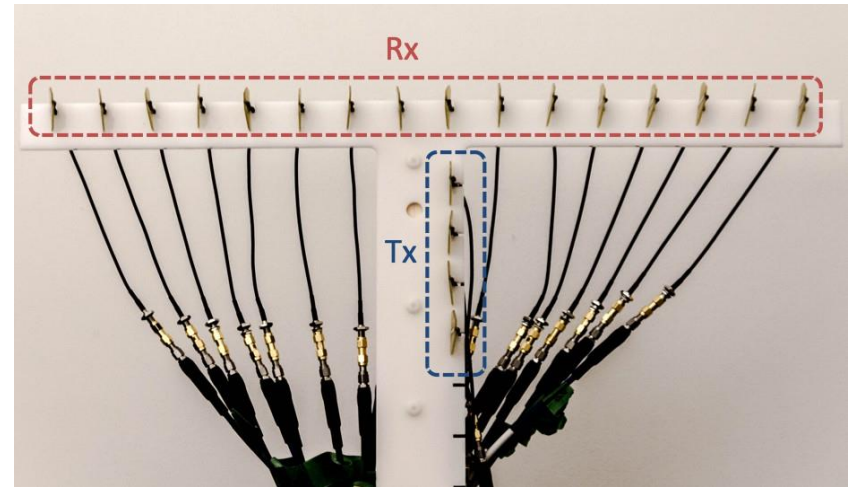
Human Walks toward Sensor



Implementation

- Hardware

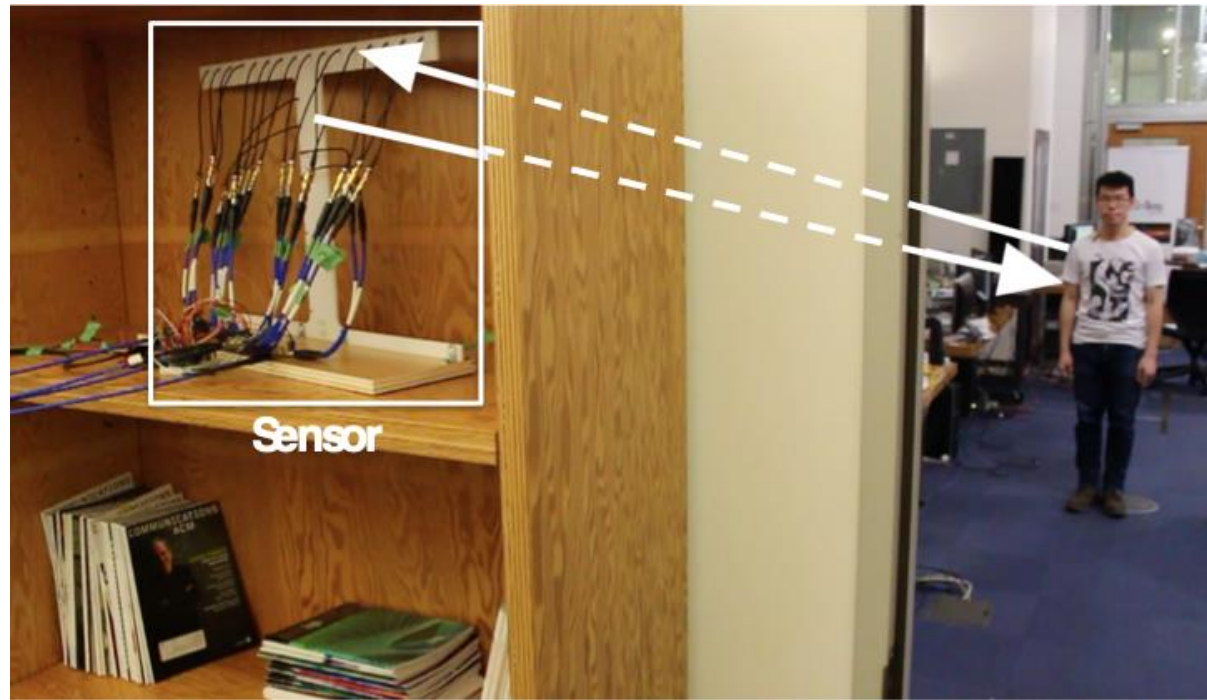
- 2D Antenna Array
- Built RF circuit
 - 1/1,000 power of WiFi
 - USB connection to PC



- Software

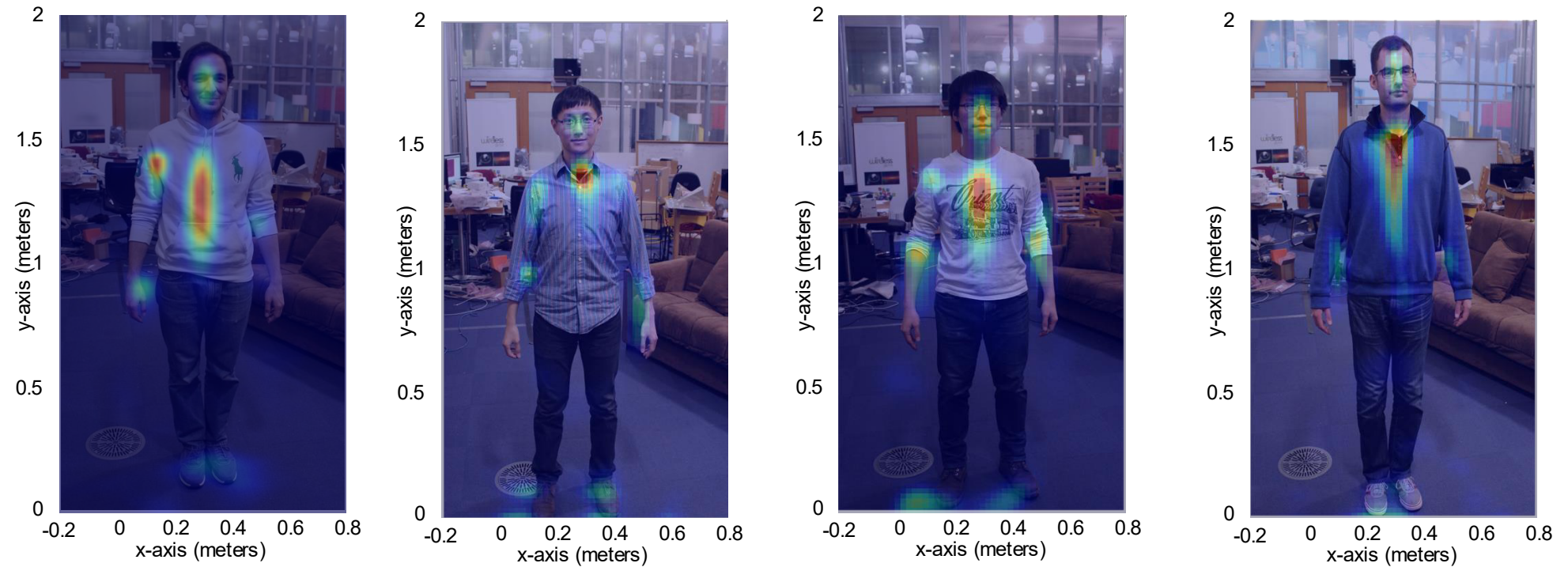
- Coarse-to-fine algorithm implemented in GPU to generate reflection snapshots in real-time

Evaluation

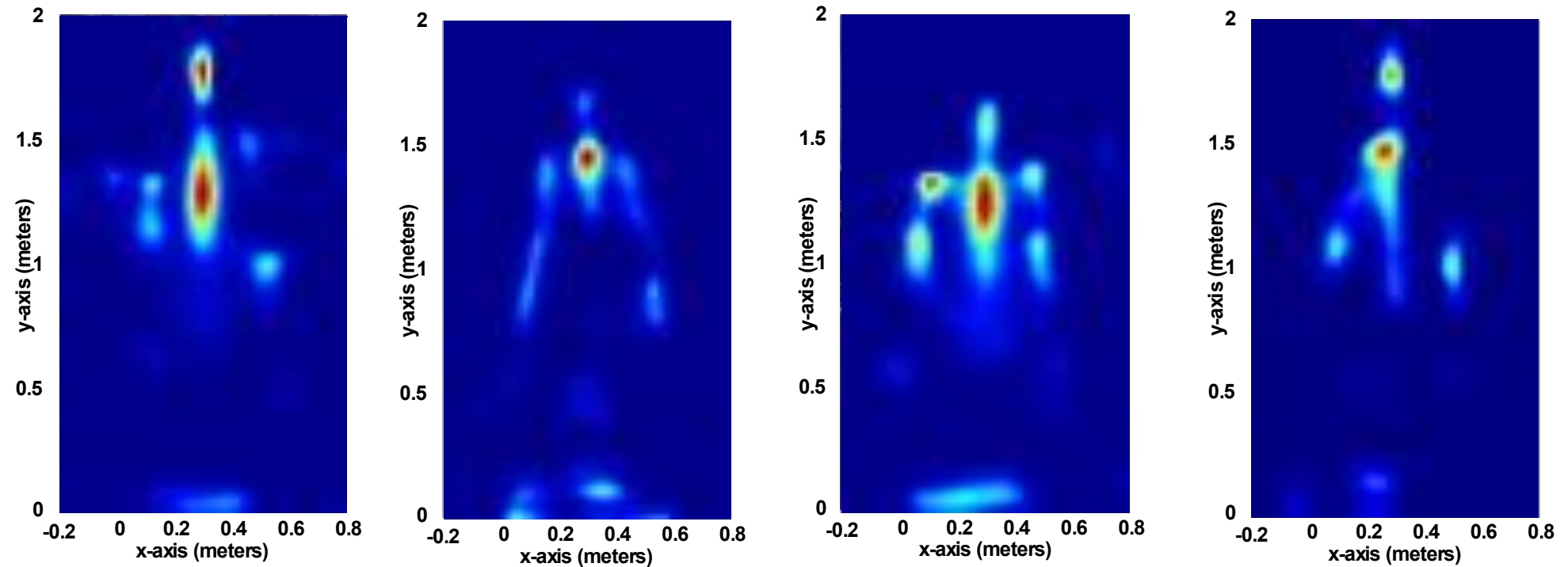


- RF-Capture sensor placed behind the wall
- 15 participants
- Use Kinect as baseline when needed

Sample Captured Figures through Walls



Through-wall classification accuracy of 90% among 13 users



Can you tell people's emotions even if they don't show up on their faces?

Smart Homes that adapt to our mood



Did I get the Job? No



Does my advisor like my work?

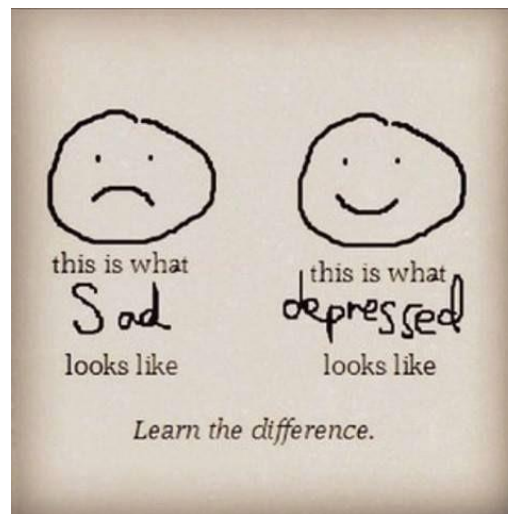


Graduate student



Advisor

Combating Depression

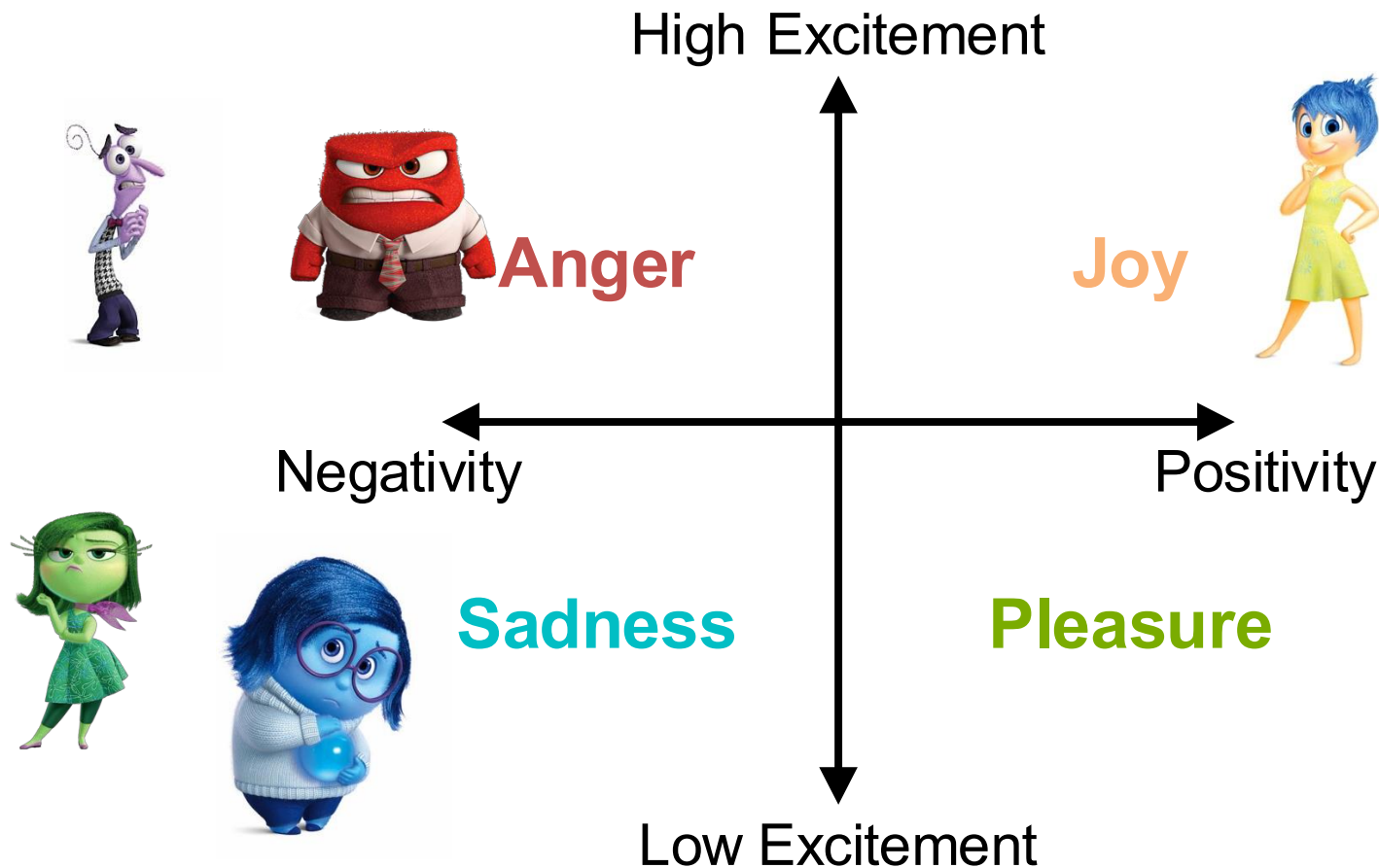


Is the date going well!

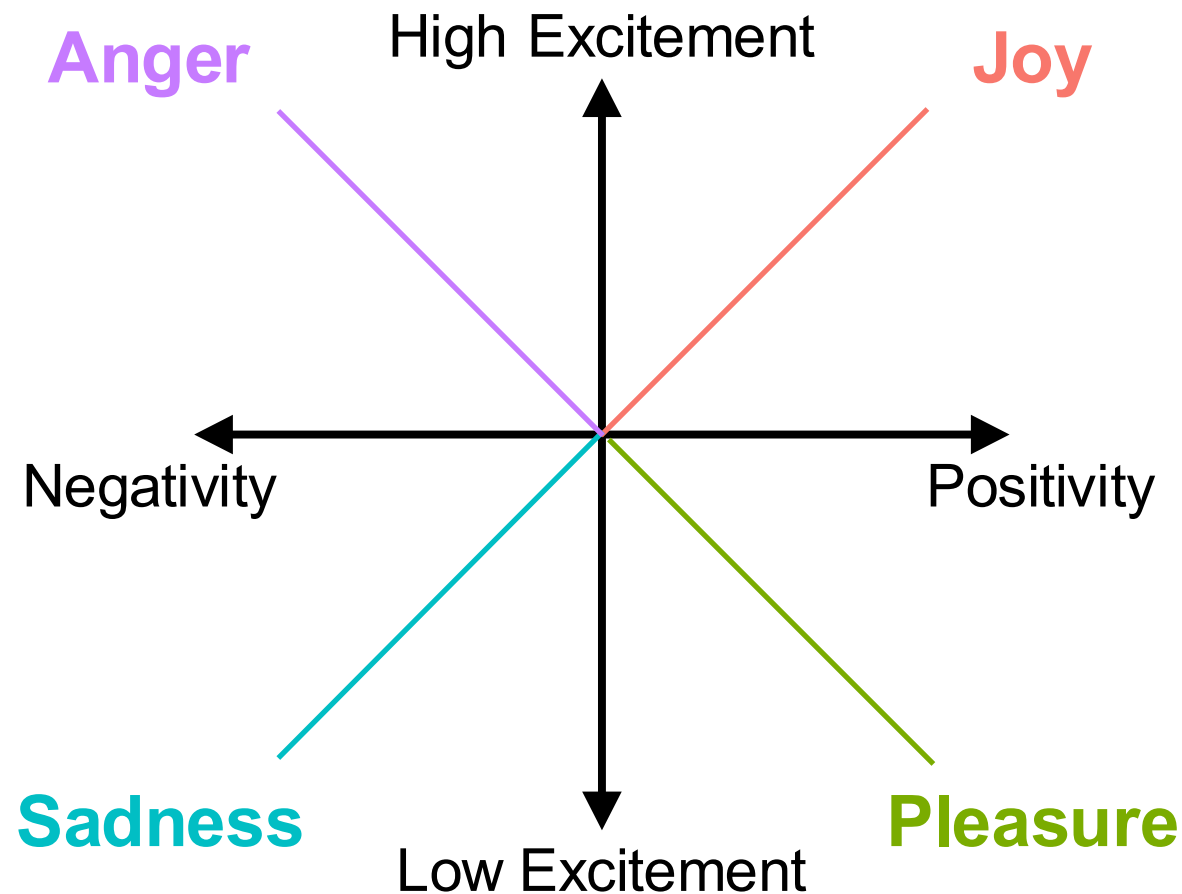


Emotion Model

- Standard 2D emotion model
- Classify into **anger**, **sadness**, **pleasure** and **joy**

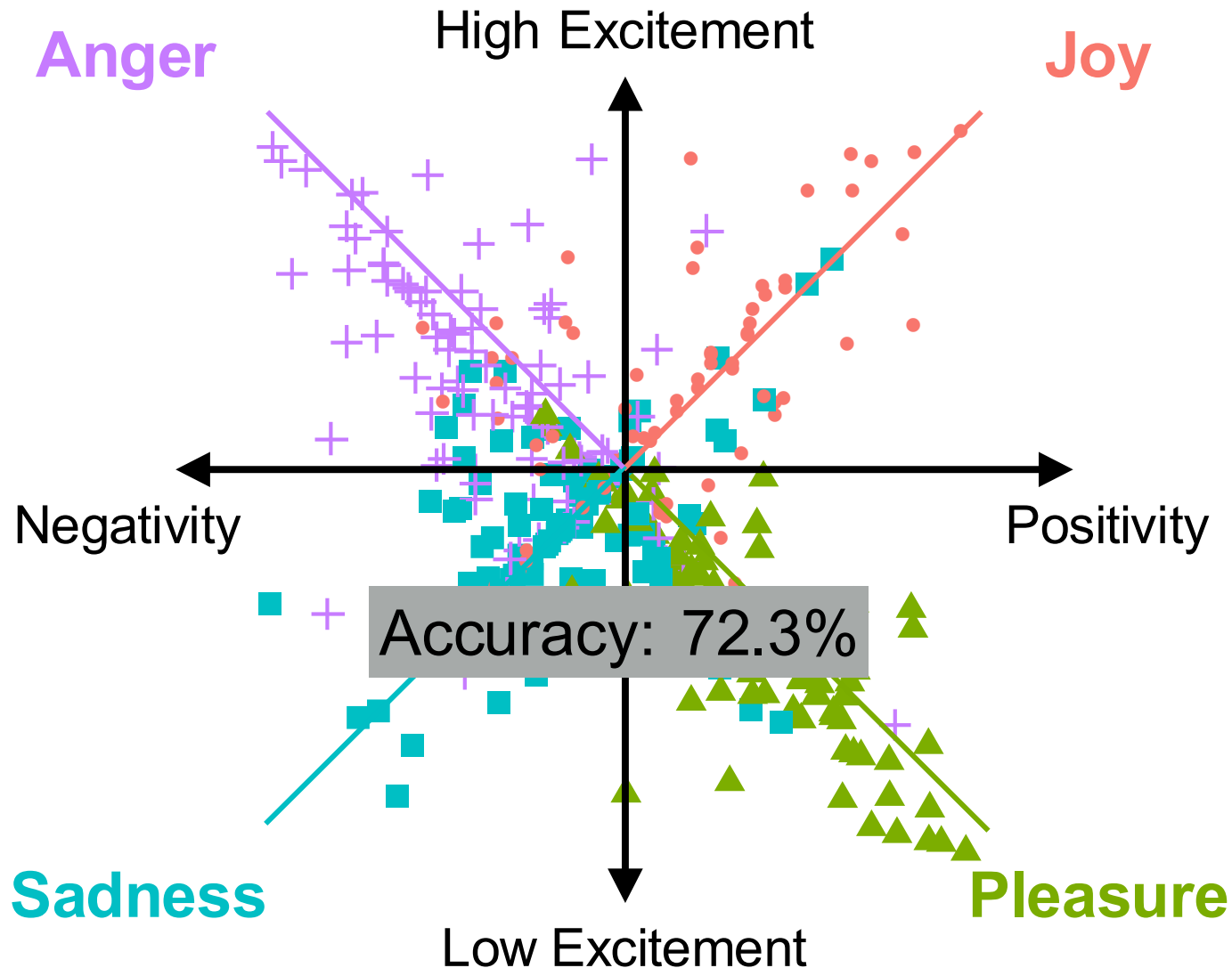


Does it detect emotion accurately?



Person-independent Classification

- Train and test on the different person



Wireless Sensing Sleep Stages

